

Mega Firms and New Technological Trajectories in the U.S.*

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August 13, 2026

Abstract

We provide evidence that mega firms have played an increasingly important role in shaping new technological trajectories in recent years. The share of novel patents—defined as patents introducing new combinations of technological components—filed by mega firms declined until around 2000 but has rebounded sharply since then. We further show that mega firms’ novel patents have generated greater technological impact after 2001 than those filed by non-mega firms. While non-novel patenting by mega firms is positively associated with subsequent growth in their market shares, novel patenting is not, suggesting that mega firms’ novel innovations may benefit the broader economy rather than primarily serve to entrench their market positions. We also explore potential correlates of this trend, showing that the rise in novel patenting by mega firms is associated with their disproportionate increase in cash holdings and the expansion of their technological scope. Our findings highlight an overlooked positive role of mega firms in the economywide innovation process.

Keywords: Mega Firms, Innovation, Novel Patents, Knowledge Diffusion

JEL Codes: O31, O33, O34, L11, L25

*Any opinions and conclusions expressed herein are those of the authors and do not necessarily represent the views of the World Bank, the Federal Reserve Board of Governors, or its staff. Previously circulated under the title “Mega Firms in Recent Trends in the U.S. Innovation: Empirical Evidence from the U.S. Patent Data.”

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1 Introduction

The concentration of economic activities in the largest businesses, so-called mega firms, has been increasing over the past few decades (Autor et al., 2020; Yeh et al., 2022; Hsieh and Rossi-Hansberg, 2023; Kwon et al., 2024).¹ In exploring the causes and consequences of this trend, a key topic of interest has been the role of mega firms in innovation and knowledge diffusion. Several studies raise concerns that mega firms may be strategically building patent thickets to hinder technological competition (Akcigit and Ates, 2023), reallocating inventors toward large incumbents (Akcigit and Goldschlag, 2023), or even acquiring innovative startups to discontinue competing R&D projects to preempt future competition (Cunningham et al., 2021). Taken together, this evidence raises the possibility that mega firms may be slowing the pace of innovation and limiting the spread of technological advances across the broader economy.

Against the backdrop of concerns about the negative impact of mega firms, this paper presents new evidence that highlights a different and understudied role of mega firms: their growing contribution to pioneering new technological trajectories that open up opportunities for subsequent innovation by other firms. To this end, we employ the concept of *novel patents*, defined as patents that introduce new combinations of technological components that had not previously been used together (Fleming, 2001; Akcigit et al., 2013; Strumsky and Lobo, 2015; Verhoeven et al., 2016; Pezzoni et al., 2022). This definition aligns with Schumpeter’s (1911) view of innovation as new combinations of existing resources (Akcigit et al., 2013) and Weitzman’s (1998) characterization of innovation as a recombination process. While prior studies show that novel patents are more likely to be antecedents of radical breakthrough (Verhoeven et al., 2016), we provide extensive evidence that they are also strongly correlated

¹Recent literature explores two broad sets of interpretations for the drivers of the rise in mega firms. Some studies have emphasized the rise in market power (De Loecker et al., 2020), possibly driven by increasing entry barriers, regulation, and lobbying activities that stifle competition (Gutiérrez and Philippon, 2019; Covarrubias et al., 2020). Other studies have cast doubt on this interpretation (Foster et al., 2022) and instead emphasize increased competition or winner-takes-all dynamics caused by globalization and technological advances that enable productive firms to exploit economies of scale (Autor et al., 2020; Hsieh and Rossi-Hansberg, 2023).

with a range of indicators of novelty and technological impact (e.g., Kogan et al., 2017; Akcigit and Kerr, 2018; Arts et al., 2021; Kelly et al., 2021; Kalyani, 2024; Jo and Kim, 2026).

To fix ideas, we illustrate an example of a novel patent. U.S. patent No. 8,188,868 titled “Systems for activating and/or authenticating electronic devices for operation with apparel” was filed in 2006 by Nike, which combined CPC group G08C17 with CPC groups A43B3 and A41D1 for the first time.² This technology integrates wireless transmitting devices into apparel—such as T-shirts and shoes—to allow athletes to monitor vital signs and performance. It was initially commercialized in 2006 through a collaboration with Apple, resulting in NIKE+iPod Sports Kit. While the extent to which this technology could create a new market or technological domain was highly uncertain at the time, ex-post evaluation suggests it laid the groundwork for subsequent innovations in wearable technology, motion sensing, and integrated digital services—including Apple Watch, Fitbit, and Nintendo Wii. A visual depiction of the patent is presented in Figure A1 in the Online Appendix.³

Our main finding is that while the share of novel patents filed by mega firms in the total number of novel patents had been on the decline throughout the 1980s and 1990s, this trend reversed sharply around 2001. Since then, mega firms’ contribution to novel innovation has been on the rise and by the mid-2010s, their share in total novel patent applications had risen to its highest level since the beginning of our sample in 1980. To assess the robustness of this finding, we estimate firm-level panel regressions and show that, after 2001, mega firms became more likely than non-mega firms to file novel patents, even after controlling for firms’ total number of patent applications, firm size, and any time-invariant characteristics. This suggests that the trend reversal noted above is not driven solely by mega firms becoming larger or filing more patents overall. We also use an alternative text-

²CPC stands for Cooperative Patent Classification, a system developed by the U.S. Patent and Trade-mark Office (USPTO). G08C17 is “Arrangements for transmitting signals characterized by the use of a wireless electrical link,” A43B3 is “Footwear characterized by the shape or the use” and A41D1 is “Garments.”

³Several additional recent examples are discussed in Section 5.3.

based measure of “breakthrough patents” from Kelly et al. (2021) and find that mega firms’ share of breakthrough patent filings has likewise increased significantly since the early 2000s. This reinforces our finding by demonstrating that mega firms are not only generating more novel patents, but also accounting for a growing share of the most influential technological advancements.

Importantly, the rebound in the share of novel patents filed by mega firms coincided with a broader rebound in the share of novel patents among all U.S. patent applications. Although the absolute number of novel patent applications remained relatively steady during the 1990s and early 2000s, their share declined due to the rapid increase in total patent applications. Between 2005 and 2016, however, the number of novel patent applications more than doubled, and by 2016, their share in total patent applications reached levels not seen since 1995. We also find that while the increase in the overall number of patents by mega firms was associated with subsequent increase in their market share (thus, suggesting potential strategic use of patenting as surmised by the extant literature), we find no such association between increased novel patents by mega firms and their market shares.

To further evaluate the technological impact of novel patents by mega firms, we adopt a measure used in prior studies (e.g., Pezzoni et al., 2022), which counts the number of “follow-on patents”—patents that reuse the same new technological combination originally introduced by a novel patent. We find that after (and only after) 2001, novel patents filed by mega firms on average generate more follow-on patents and are more likely to become *hits*—defined as patents that give rise to an exceptionally large number of follow-on patents and thus significantly shape subsequent technological trajectories—than those filed by non-mega firms. These results are robust to using forward citation counts, a more traditional measure of technological impact.

One straightforward way to examine the degree of knowledge diffusion stemming from novel patents is to analyze the extent to which follow-on patents are produced outside the firm that originally generated the novel patent. We find that “self-follow-on rate”—the

proportion of follow-on patents generated by the same firm that filed the original novel patent—is similar for mega firms and non-mega firms prior to 2001. However, after 2001, we find some evidence of a slowdown in knowledge diffusion from novel patents produced by non-mega firms, whereas the trend remains stable for mega firms.⁴ Together with mega firms’ novel patents generating more follow-on patents in recent years, this finding suggests that mega firms contribute to knowledge diffusion beyond their boundaries by engaging in technological experiments and generating impactful new combinations, a channel that has been understudied in the literature.⁵

An important question is what factors could potentially explain the rise in novel innovation by mega firms since the early 2000s. We propose two hypotheses and explore the supporting empirical evidence. First, we hypothesize that, given the highly experimental and uncertain nature of novel innovation, firms with larger cash holdings might be better positioned to undertake such innovation, and that mega firms have been increasingly holding more cash relative to non-mega firms since the early 2000s. This hypothesis aligns with findings in the finance literature showing that larger cash holdings facilitate novel, riskier innovation (Krieger et al., 2022), and that U.S. firms have been increasing their cash holdings in part to fund risky R&D investments (Bates et al., 2009; Brown et al., 2009).

We find empirical evidence consistent with this hypothesis, although the evidence should be interpreted as correlational rather than causal. To begin with, firms with larger cash holdings are indeed more likely to engage in novel innovation. Using a local projection method, we show that a one percent increase in cash holdings is associated with a gradual rise in novel patent applications, reaching a 6.2 percent increase after four years. Total patent applications also increase, but the magnitude of the association is less than half as large as for novel patents. This pattern suggests that cash holdings are more strongly correlated with

⁴These results are broadly consistent when using self-forward citation rates, a commonly used proxy for knowledge diffusion in the literature (e.g., Akcigit and Kerr, 2018; Jo and Kim, 2026).

⁵Patent reassignment and acquisitions may be another way for mega firms to defend their technological leadership (Akcigit and Ates, 2023). Subsequent changes in patent ownership are outside the scope of our analysis as we focus on the initial applicants for novel patents.

novel innovation than with patenting activity in general.

Examining aggregate trends, we find that the gap in cash holdings between mega and non-mega firms exhibits a U-shaped pattern, with a reversal in the late 1990s. This gap narrowed by as much as 50% of its 1980 value through the late 1990s, but experienced a strong rebound thereafter. By 2016, the final year of our sample, the difference in cash holdings between mega- and non-mega firms had exceeded the 1980 level by 15% to 30%. Taken together, these findings support the idea that the rise in cash holdings among mega firms after 2001 may have facilitated their engagement in novel innovation.

Second, we note that novel innovation involves recombining existing knowledge, and as Weitzman (1998) shows, the probability of generating new combinations increases with the breadth of the existing knowledge base. As the stock of knowledge grows, achieving novel breakthroughs often requires not only larger but more diverse teams of inventors with increasingly specialized expertise (Jones, 2009; Freund, 2025). Building on this idea, we examine trends in the relative technological scope (i.e., diversity of expertise) of inventor teams at mega firms compared to non-mega firms, controlling for team size. We find that mega firms experienced a relative decline in the technological scope of their inventor teams during the first two decades of our sample, but this trend was reversed in more recent years. This suggests that an expansion in technological scope—rather than sheer inventor team size—may have given mega firms a new competitive advantage in producing novel patents, consistent with the definition of novel patents as those combining technological components that had not been previously used together.

Our findings have important policy implications. If it is true that mega firms are stifling innovation and slowing down knowledge diffusion, there may be scope for regulatory intervention. If, however, those firms are among the key actors conducting experiments and generating new technological trajectories, then such an approach may backfire. With the U.S. technological dominance facing increasing global challenges, the stakes could not be higher. We provide further discussion in the concluding section.

Relation to the Literature Our paper engages with the ongoing debate surrounding the rise of mega firms and its implications for market dynamics and innovation. While prior research has extensively documented the increasing concentration of economic activities in large firms (Autor et al., 2020; Yeh et al., 2022; Hsieh and Rossi-Hansberg, 2023) and proposed competing explanations—ranging from increased market power (De Loecker et al., 2020; Gutiérrez and Philippon, 2019; Covarrubias et al., 2020) to intensified competition driven by globalization and technology (Autor et al., 2020; Hsieh and Rossi-Hansberg, 2023; Kwon et al., 2024; Foster et al., 2022)—our study shifts the focus to the role of mega firms in innovation, particularly through the lens of novel patents. By analyzing mega firms’ contributions to novel technological combinations, our study sheds new light on whether their growing influence enhances or hinders technological progress and whether their market position is driven by innovative capabilities or by market power that may limit competition.

Our paper is also related to the literature on innovation which has recently raised concerns over declining efficiency of R&D investment (Bloom et al., 2020) and the general declining trend in the U.S. innovation creativity (Kalyani, 2024). Since this trend and the aforementioned trend toward increasing dominance of mega firms happened concurrently in time, this has led to the examination of the interrelationship between the two trends, with much of the literature focusing on the stifling effect of the rise of mega firms on innovation through strategic patenting and a slowdown in knowledge diffusion (Akcigit and Ates, 2023), increasing concentration of inventors in the largest firms (Akcigit and Goldschlag, 2023), and “killer acquisitions” (Cunningham et al., 2021). Some more recent studies suggest, however, that part of the advantages of mega firms may actually arise from the nature of technology (Gupta et al., 2024). Our main contributions to this body of literature lie in (re-)examining the role of mega firms in innovation through the prism of novel patents and over several decades.

Our paper also builds upon the strand of the literature which has documented growing importance of new combinations of technologies in innovation and economic growth. This

literature has examined the technological origins of novel patents that create new knowledge combinations using various measures, most prominently, technological classes assigned to patents by patent examiners (Fleming, 2001; Strumsky and Lobo, 2015; Verhoeven et al., 2016; Epicoco et al., 2022). Akcigit et al. (2013) present broad historical evidence showing the growing importance of new technological combinations (novel patents) over a century and a half in the U.S. innovation. The most closely related paper is Pezzoni et al. (2022) which also looks at follow-on patents (trajectories of new combinations). Additionally, recent studies have leveraged natural language processing (NLP) techniques to develop more nuanced measures of patent novelty and impact, such as those by Arts et al. (2021), Kelly et al. (2021), Kalyani (2024), and Kalyani et al. (2025). These advancements provide valuable benchmarks for assessing the technological significance of patents, which we utilize to validate and compare to our measure of novel patents. Most of the extant papers, however, are limited to examining the technological novelty and the role of recombined knowledge in patents’ impact and do not look into who creates new combinations. We link this literature to the above literature strand on the role of mega firms to examine how much such firms contribute to novel patents and what type of new potential technological trajectories they generate to better understand their changing role in the economy-wide innovation process.

The remainder of the paper is structured as follows. Section 2 describes the data and key measurement methods. Section 3 examines the aggregate trend in mega firms’ contributions to novel innovation, along with firm-level evidence. Section 4 analyzes the technological impact and knowledge diffusion of novel patents by mega firms compared to non-mega firms over time. Section 5 explores potential drivers of the rise in novel patents by mega firms. Finally, Section 6 concludes.

2 Data and Measurement

2.1 Data Construction and Key Measurement Methods

The primary data sources are the USPTO PatentsView and Compustat. The USPTO PatentsView tracks all patents ultimately granted by the USPTO from 1976 onward. We collect utility patents granted to U.S. assignees between 1976 and 2023 to track economy-wide innovation activities, and in particular, the creation and trajectories of new technological combinations. We describe detailed matching procedures in the Online Appendix [A.2](#).⁶

To identify technological components underlying an invention, we exploit the detailed information provided by the USPTO patent database on the technological content of inventions. Each patent documentation in the USPTO reports technology classes based on all disclosed information. To conduct an efficient patent search, the USPTO requires patent examiners to objectively classify an invention into technology categories based on “invention information” and “additional information.” In this paper, we use technology classes based on “invention information,” which, according to the USPTO, contains “technical information in the total disclosure of a patent document (for example, description, drawings, claims) that represents an addition to the state of the art.”

We utilize the Cooperative Patent Classification (CPC) to identify technological components of inventions. The CPC scheme is a hierarchical system with multiple levels of classifications. The level of classification we use in this paper is “Main Group”—the most comparable level of classification to the USPC subclass widely used in the previous literature.⁷ Hereafter, we use “technological components” and “main group” interchangeably. While new technological components are added over time, the USPTO reclassifies old patents according to the new CPC code, which ensures comparability over time. By 2016, there were 7,246 distinct main groups under the CPC scheme excluding those under CPC Section Y.⁸

⁶While we use our own USPTO-Compustat bridge in this paper, we confirmed that our basic findings are robust to using the DISCERN (Duke Innovation & Scientific Enterprises Research Network) bridge.

⁷Our findings are robust to using different levels of aggregation as well as the IPC classification.

⁸Section Y represents a new addition to patent classifications introduced together with CPC, for general

Following previous studies (Fleming, 2001; Akcigit et al., 2013; Strumsky and Lobo, 2015; Verhoeven et al., 2016), we define a new technological combination as a pairwise combination of technological components that appears in a patent for the first time. Patents incorporating such new technological combinations are defined as novel patents. While our analysis is based on utility patents assigned to the U.S. entities, we identify a pair of technological components as a new combination only if it is the first combination that appears among all utility patents granted to both U.S. and non-U.S. entities since 1976. Because the first year available in the USPTO PatentsView data is 1976, we do not observe the history of technological combinations created before then. We use the first four years, 1976-1979, as a buffer period and we track novel combinations starting from 1980.

Note that, by construction, a novel patent may contain multiple new pairwise technological components. While novel patents rely on new pairwise combinations of technological components, we aggregate them to the patent level and use novel patents as our measure of novel innovation, in particular because the number of pairwise new combinations is more likely to overestimate the true number of novel innovations than the measure aggregated to the patent level. Nevertheless, we have verified that all our findings remain robust to using new pairwise combinations as the measure of novel innovation.

To study the diffusion and technological trajectories of new combinations, as well as to develop a measure of novel patents' impact, we identify the pool of follow-on inventions of a novel patent as subsequent patents that (re-)use the same combination of technological components as introduced by the novel patent. Specifically, we count the cumulative number of patented inventions that re-use the new technological combination in years following the appearance of the focal new combination, up to late 2023. Furthermore, to gauge the extent to which subsequent innovation occurs beyond the boundaries of the focal firm that generated the novel patent, we differentiate follow-on patents that are generated by the focal firm versus those generated by other firms. Occasionally, patents are assigned to multiple assignees. In

tagging of new technological developments which are already classified or indexed in other sections. We exclude technological components tagged under this section when constructing new combinations.

such cases, we say a follow-on patent is generated by the focal firm if it is one of the assignees.

2.2 Validation of Measurement

While the concept of new technological combinations and the measurement of novel patents based on this idea have been developed in the literature since at least 2007, more recent studies incorporating Natural Language Processing (NLP) techniques have introduced alternative measures of patent novelty and impact. In this section, we briefly examine the relationship between the novel patent measure used in our paper and these alternative measures. Later, we discuss how these measures capture the contributions of mega firms compared to our approach.

Specifically, we first investigate the relationship between our measure of novel patents and three alternative measures. The first two are both based on NLP techniques. The first measure, developed by Arts et al. (2021), extracts the number of new bigrams—compared to all patents filed since 1969—from the title, abstract, and claims of a patent. The second measure is the patent creativity measure due to Kalyani (2024), which is defined as the share of creative technical bigrams—where creativity is determined by these bigrams not appearing in any patent filed in the previous five years—relative to all technical bigrams. The third measure we investigate is backward self-citation ratios, defined as the number of citations to a firm’s own previous patents divided by the total number of citations. The endogenous growth literature (e.g., Akcigit and Kerr, 2018; Jo and Kim, 2026) often uses this variable to capture innovation associated with creative destruction—where firms pursue new ideas that expand beyond their existing technological scope to replace incumbents in other markets. A higher self-citation ratio is typically linked to incremental improvements in a firm’s existing technology, whereas a lower self-citation ratio suggests a shift toward novel areas.

In Table 1 we use patent-level data from 1980 to 2016 to estimate the relationship between our novel patents measure and those alternative measures. Specifically, we estimate regressions where the dependent variables capture various aspects of patent novelty and

creative destruction, and the main independent variable is our novel patent indicator. The results in Column (1) show a strong positive relationship between our measure and that in Arts et al. (2021): we find that while non-novel patents have 1.5 new bigrams on average, novel patents have an additional 0.8 new bigrams.⁹ Similarly, the results in Column (2) indicate a strong positive relationship between our measure and the patent creativity measure in Kalyani (2024). Lastly, the results in Column (3) show that novel patents are associated with lower backward self-citation ratios. All regressions include CPC group fixed effects and application year fixed effects, but the results remain robust to their exclusion or to alternative specifications with different combinations of fixed effects, as shown in Online Appendix A.3.

Table 1: Novel Patents and Other Measures of Novelty and Creative Destruction

	(1)	(2)	(3)
	# New Bigrams	Patent Creativity	Backward Self-citation Ratio
Novel Patent Indicator	0.769*** (0.019)	0.020*** (0.000)	-0.027*** (0.001)
CPC Group FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Mean Dep. Var. (Non-novel)	1.516	0.068	0.121
Obs.	2,630,141	2,575,446	2,657,952
R-sq	0.06	0.12	0.06

Notes: The estimates are obtained from OLS regressions with multi-way fixed effects. “# New Bigrams” indicates the number of new bigrams in patent text, obtained from Arts et al. (2021) and “Patent Creativity” is the share of new technical bigrams in patents, obtained from Kalyani (2024). Backward self-citation ratio is defined as the number of citations to a firm’s own previous patents divided by the total number of citations. The regression estimates are derived from all patents that were applied between 1980 and 2016. Mean Dep. Var. (Non-novel) is the mean of each outcome for non-novel patents, provided to aid interpretation of the estimated coefficients. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In the following sections, we will discuss the role of mega firms in generating novel patents as well as the impact their patents have on follow-on patents. Anticipating this, in Table 2 we also present the results of estimating the relationship between our measure of novel patents and several other measures of patents’ impact. Specifically, we again estimate regressions where the dependent variables capture various measures of a patent’s impact while the main

⁹Arts et al. (2021) also extract the number of new words and new trigrams in a patent. The regression coefficients remain similarly large and positive when using these alternative measures. They also develop a measure of backward cosine similarity, for which no meaningful difference is observed.

independent variable remains our novel patent indicator. Column (1) uses the stock market valuation of patents (expressed as the log of real dollar values) from Kogan et al. (2017), which reflects the patent’s contribution to a firm’s current and future profits. The results indicate a modest but positive effect, with novel patents valued approximately 2% higher than non-novel patents. Column (2) examines five-year forward citations, the most widely used measure of scientific impact. On average, novel patents receive 0.7 more citations than non-novel patents, which receive an average of 5.5 citations.

Lastly, Column (3) utilizes the “breakthrough” patent indicator from Kelly et al. (2021). This measure combines a measure of patent novelty with a measure of its impact. As such, Kelly et al. (2021)’s measure is conceptually most closely related to the discussion of novel patents by mega firms and their impact in Sections 3 and 4 below, while being constructed in a totally different way. In constructing their measure, Kelly et al. (2021) also leverage NLP techniques to first quantify textual similarity between patent pairs, identifying a patent as novel if its content is distinct from prior patents and impactful if it is similar to future patents. They then compute the ratio of impact to backward similarity (i.e., the inverse of novelty), where higher values indicate distinct advancements at the technological frontier that serve as a foundation for subsequent inventions. The top 10% of patents in the resulting importance distribution are classified as breakthrough patents.¹⁰ The estimation results show that novel patents are 1.5 percentage points more likely to be breakthrough patents than non-novel patents, which have a 10% probability of being classified as breakthrough.¹¹

¹⁰Specifically, Kelly et al. (2021) define their measure as the similarity of a patent to all patents filed over the next τ years, divided by its similarity to all patents filed in the previous five years. Their baseline analysis uses $\tau = 10$, though they also construct similar measures for $\tau = 5$ and $\tau = 1$. In our analysis, we use $\tau = 5$ to ensure the measure can be calculated for patents filed through 2016.

¹¹We also examine how the results in Table 2 vary under different combinations of CPC group and year fixed effects. Controlling for year fixed effects is essential for revealing the positive relationship in Column (1) of Table 2, as it accounts for the time trend in the overall stock prices that are driven by macroeconomic factors such as changes in long-term interest rates or equity risk premium. Similarly, controlling for CPC group fixed effects is necessary to uncover the positive relationships in Columns (2) and (3), as it adjusts for systematic differences in the number of patents across technology groups. Further details are provided in Online Appendix A.4.

Table 2: Impact of Novel Patents

	(1)	(2)	(3)
	Market Valuation	5yr Forward Citation	Breakthrough
Novel Patent Indicator	0.021*** (0.005)	0.690*** (0.041)	0.015*** (0.001)
CPC Group FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Mean Dep. Var. (Non-novel)	1.825	5.482	0.102
Obs.	1,504,550	2,838,910	2,836,755
R-sq	0.09	0.04	0.16

Notes: The estimates are obtained from OLS regressions with multi-way fixed effects. “Market Valuation” is the stock market value of a patent obtained from Kogan et al. (2017), and “Breakthrough” is an indicator of whether a patent is a breakthrough patent, obtained from Kelly et al. (2021). The regression estimates are derived from all patents that were applied between 1980 and 2016. Mean Dep. Var. (Non-novel) is the mean of each outcome for non-novel patents, provided to aid interpretation of the estimated coefficients. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

3 Contribution of Mega Firms to Novel Innovation

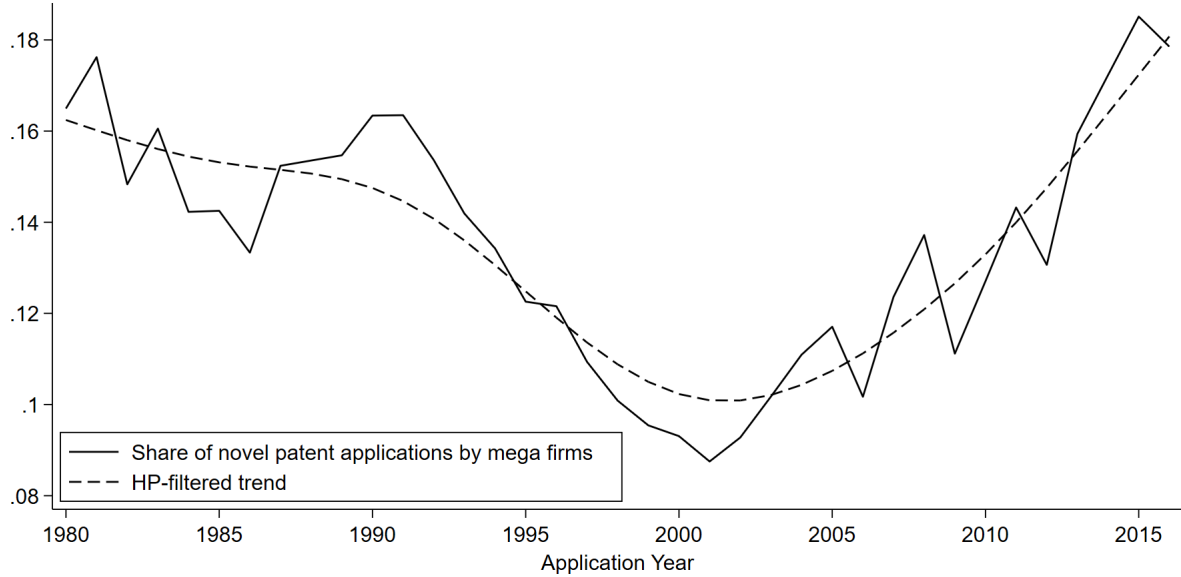
3.1 Aggregate Trend

In our baseline specification, we define mega firms as the top 50 firms by sales in each year among all publicly listed firms in the Compustat dataset. We show in robustness analyses below that our results are robust to alternative definitions of mega firms. Although publicly listed firms represent a selected subset of all firms, this approach is appropriate for our purpose of identifying the largest firms in the economy, because most of the largest U.S. firms are publicly traded. While these 50 firms comprise only a tiny fraction of the millions of firms in the U.S., our calculations indicate that, on average, they account for 8% of employment, 14% of sales, and 17% of patent applications annually in the U.S. economy.¹² Thus, they represent a sufficiently large share of business activity and innovation to have a meaningful impact at the aggregate level.

Figure 1 shows the dynamics of the contribution of mega firms to novel innovation. The

¹²Aggregate employment data are obtained from the Business Dynamics Statistics of the Census Bureau, and aggregate business receipts are from the Statistics of Income published by the Internal Revenue Service. Within the Compustat dataset, mega firms account for 23% of employment, 31% of sales, and 26% of patent applications.

Figure 1: Share of Novel Patent Applications By Mega Firms



Source: Authors' own calculation using the USPTO patent data matched with Compustat data.

Notes: The figure presents the share of novel patent applications filed by mega firms as a proportion of all novel patent applications from 1980 to 2016 (solid line), along with its smoothed trend estimated using the Hodrick-Prescott filter with a smoothing parameter of 100 (dashed line). Mega firms are defined as the top 50 firms by sales in Compustat for each year. Note that the vertical axis does not start at zero.

solid line indicates the share of novel patent applications by mega firms out of all novel patent applications applied by U.S. patent assignees and the dashed line is its smoothed trend. The share displays a U-shaped trend: There had been a prolonged decline from 16 percent in 1980 to nine percent in 2001, followed by a steep increase afterward reaching its historical high level of 18 percent in 2016.

The choice of the top 50 firms is not intended to imply any particular significance of that cutoff, but rather to capture the largest firms in the economy. In Online Appendix A.5, we show that the U-shaped trend remains robust if we broaden the definition of mega firms to the top 100 firms by sales each year, or the top 4 firms in each four-digit NAICS industry—an approach comparable to the “superstar firms” definition in Autor et al. (2020). The latter result indicates that the trend is *not* driven by the changes in industry composition among the largest firms in the economy—for example, the shift from industrial giants in the 1980s to technology giants in the 2010s. The trend also holds when defining mega firms as the top 50 firms by sales in each year among patenting firms only, accounting for the fact that

patenting activity is highly concentrated in certain industries and firms.^{13,14}

From an accounting identity perspective, the observed trend in Figure 1 can be driven by two forces. The rise since the early 2000s, for example, may reflect an increase in the total number of patent applications by mega firms or a greater tendency for those patents to be novel. To disentangle these forces, we decompose the share of novel patents produced by mega firms as follows:

$$\underbrace{\frac{N_{m,t}}{N_{m,t} + N_{o,t}}}_{Y_t : \text{Mega firm share (novel)}} = \underbrace{\frac{N_{m,t}/T_{m,t}}{(N_{m,t} + N_{o,t})/(T_{m,t} + T_{o,t})}}_{X_{1,t} : \text{Relative tendency}} \times \underbrace{\frac{T_{m,t}}{T_{m,t} + T_{o,t}}}_{X_{2,t} : \text{Mega firm share (total)}} \quad (1)$$

where $N_{m,t}$ and $N_{o,t}$ are the number of novel patents applied by mega firms and other firms, respectively, and $T_{m,t}$ and $T_{o,t}$ are the total number of patents applied by each group. The first term, $(X_{1,t})$, captures the relative tendency of mega firms to engage in novel innovation compared to all firms, and the second term, $(X_{2,t})$, represents the share of total patent applications accounted for by mega firms. The time series for $X_{1,t}$ and $X_{2,t}$ are presented in Online Appendix A.6.

By taking logs on both sides of equation (1) and time-differencing, we can quantify the contribution of each component to changes in the share of novel patents by mega firms over time. Between 1980 and 2000, the share of novel patents by mega firms (Y_t) declined by 7.2 percentage points, with $X_{1,t}$ and $X_{2,t}$ explaining 77 percent and 23 percent of the decline, respectively. Between 2000 and 2016, Y_t increased by 8.5 percentage points, with $X_{1,t}$ and $X_{2,t}$ explaining 35 percent and 65 percent of the increase, respectively. These results indicate that the recent rise in the share of novel patents by mega firms is driven not only by their growing share of total patent applications, but also by their increased propensity for novel

¹³This U-shaped trend is also consistent with findings of Fort et al. (2025), who show that the share of breakthrough patents filed by firms with more than 10,000 employees follows a similar U-shaped pattern.

¹⁴There are, however, a few notable industry-level exceptions to the U-shaped trend. For example, in Biochemistry and Pharmaceutical technologies, mega firms' contribution to novel innovation declined during the 1990s and did not recover thereafter (see Figure A4 in the Online Appendix). This pattern is consistent with the notion that large pharmaceutical firms increasingly rely on the acquisition of innovative smaller firms rather than undertaking novel innovation internally.

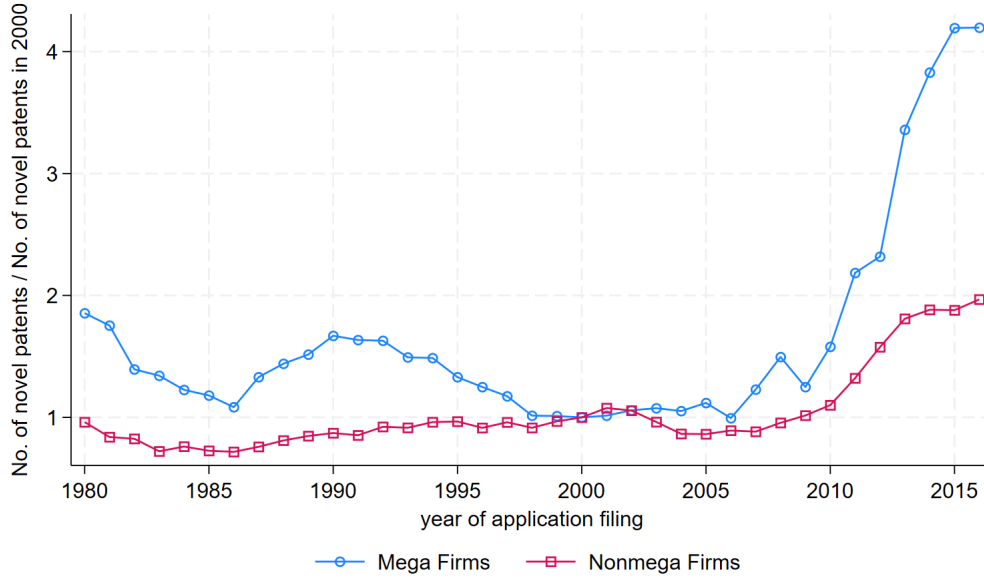
innovation, with the former playing the larger role. At the same time, it is noteworthy that mega firms substantially expanded their patenting volume after the early 2000s without a corresponding decline in novelty. To the extent that scaling up innovative activity might be expected to reduce the share of novel patents at the margin, this pattern suggests that mega firms were able to maintain a high rate of novelty even as their patenting activity grew rapidly.

Examining the number of novel patents filed by mega firms ($N_{m,t}$) and other firms ($N_{o,t}$) separately provides useful insights into the underlying trend. One possibility is that the U-shaped pattern in Figure 1 reflects a rise and fall in novel innovation by non-mega firms—perhaps driven by startup activity during the IT boom of the 1990s—while novel patenting by mega firms remained flat. However, Figure 2 below shows that this is not the case. The number of novel patents filed by mega firms (blue circles) declined between 1980 and 2000, but began rising sharply in the mid-2000s. Meanwhile, the number of novel patents filed by non-mega firms (red squares) remained relatively stable during the earlier period and also began increasing in the late 2000s, albeit to a more modest extent than mega firms. Note each series in the figure is normalized to its 2000 value for comparability between the two series.

A related concern is that our findings may simply reflect the rise of ICT innovation. Advances in information technology may have allowed large firms to expand their scale, including their patenting activity, while also creating greater opportunities for novel innovation more broadly. To examine this possibility, we assess whether the rise in mega firms’ share of novel patents since the 2000s is more pronounced in ICT fields than in non-ICT fields (see Figure A7 in the Online Appendix). We find, however, that the U-shaped pattern is present for both ICT and non-ICT novel patents, suggesting that the pattern is not simply an artifact of the broader shift toward ICT innovation.

It is worth noting that the U-shaped pattern similar to that observed for mega firms in Figure 1 above is also found in the dynamics of novel patents in the U.S. overall. As

Figure 2: The Number of Novel Patents by Mega vs. Non-mega Firms



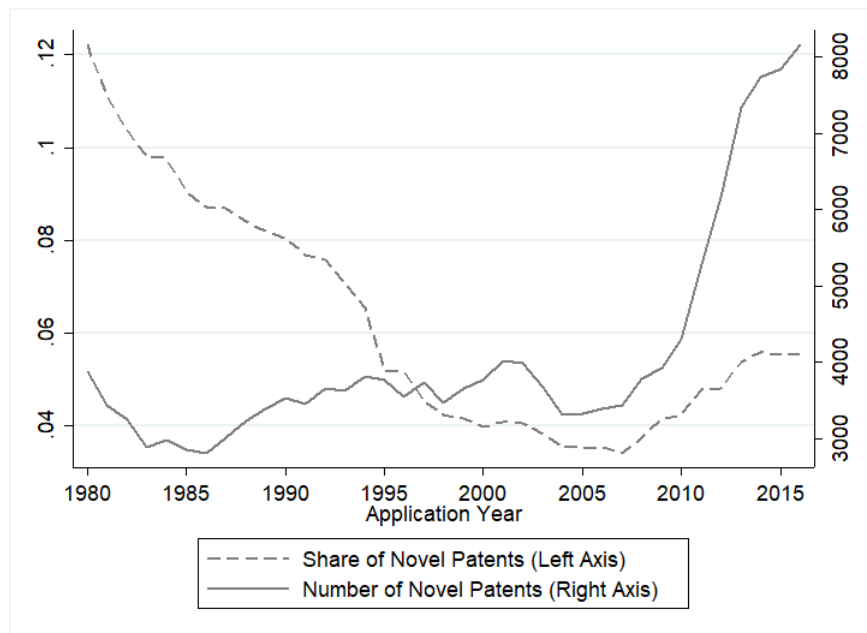
Source: Authors' own calculation using the USPTO patent data matched with Compustat data.

Notes: The figure presents the number of novel patent applications filed by mega firms (blue circle) and by non-mega firms (red square) from 1980 to 2016. Each series is normalized by its 2000 value for comparability between the two series.

mentioned, Akcigit et al. (2013) showed the long-term secular trend towards increasing importance of novel combinations in the U.S. innovation since the 1830s. However, a closer inspection of their evidence whose timeline ends in 2004 reveals that the share of novel combinations appears to have actually been declining from around the 1980s (Figure 1b in Akcigit et al. (2013)).

Consistent with this, in Figure 3 we show that the absolute number of novel patent applications assigned to U.S. entities had been basically flat, while the share of novel patents in total had been declining steadily, from 12% in 1980 to 8% at the start of the 1990s to less than 4% in the early- to mid-2000s. As the same Figure 3 shows, however, the number of novel patent applications doubled to almost 8,000 per year after the mid-2000s and their share in total patent applications had accordingly recovered to almost 6 percent, the level last seen in the mid-1990s. This renewed trend towards increasing share of novel combinations after the mid-2000s suggests that the decline in the share of novel patents from the 1980s to early 2000s may have been an aberration, after all.

Figure 3: The Number of Novel Patents and Their Share in Total Patent Applications



Source: Authors' own calculation using the USPTO patent data.

Notes: The figure presents the number of novel patent applications filed by all firms (solid line) and the share of patents that are novel (dashed line) from 1980 to 2016.

Furthermore, comparing Figures 1 and 3, we can see that the increase in the share of novel patents in total patent applications by mega firms started increasing already from around 2001, several years before the reversal of the overall trend, while the number of novel patents filed by mega firms has also increased more sharply than the number of novel patents filed by non-mega firms (Figure 2).

As the economy evolves, the set of the largest firms changes over time; the largest firms in the 1980s are not the same as those in the 2000s. This raises the question of whether the trend in novel patenting activity by mega firms is driven, at least in part, by changes in their composition over time.¹⁵ To address this question, we decompose the annual change in the number of novel patents filed by mega firms, $N_{m,t}$, into two components: one attributable to firms that remain in the top 50 from year $t - 1$ to t (continuers), and the other due to turnover in composition—i.e., entrants (newly included mega firms) and exiters (firms that

¹⁵Recall that our results are robust to defining mega firms as the top four firms within each four-digit NAICS industry, as shown in Figure A3. This indicates that the U-shaped trend remains robust when holding the industry composition constant.

drop out of the top 50).

Specifically, we decompose the annual growth rate of novel patents filed by mega firms as

$$g_t = g_t^{cont} + g_t^{comp}, \quad (2)$$

where g_t^{cont} captures the contribution from continuing mega firms, and g_t^{comp} captures the net effect of firm turnover. A detailed derivation of this decomposition, along with a visual depiction of each component, is provided in Online Appendix A.8.¹⁶ Applying this method, we find that for the vast majority of the 1980-2016 period, the year-to-year growth in novel patent applications by mega firms is predominantly driven by continuers rather than by the change in composition. Using the Shapley-Shorrocks variance decomposition to Equation (2), we find that g_t^{cont} accounts for 74% of the variation in g_t , while g_t^{comp} explains the remaining 26%.

3.2 Firm-level Evidence

We next employ a firm-year level panel regression framework to assess how robust the increase in novel innovation by mega firms since 2000 is when controlling for various firm characteristics.

To do so, we estimate a set of Poisson regressions of the following form:

$$E(Y_{it}|X_{it}) = \lambda_{it} = \exp(\beta_1 \cdot Mega_{i,t} + \beta_2 \cdot Mega_{i,t} \times Post_t + X'_{it}\gamma + \alpha_i + \eta_t), \quad (3)$$

where the dependent variable Y_{it} is the number of novel patent applications by firm i in year t , $Mega_{i,t}$ is an indicator whether firm i is a mega firm in t , $Post_t$ is an indicator whether $t \geq 2001$, X_{it} is a vector of firm-level characteristics, α_i is a firm fixed effect, and η_t is an industry-by-year fixed effect (industry notation omitted). The independent variable of

¹⁶The decomposition method is similar to that developed by Sivadasan et al. (2025) and bears some similarity to Foster et al. (2001), with a focus on decomposing growth rates rather than changes in averages.

interest is β_2 . Naturally, this regression framework uses a subset of patent assignees that are publicly-listed firms (i.e., excludes privately-owned firms and research institutions) in order to measure firm characteristics, and it compares these firms' novel patenting behavior between the 1980-2000 and 2001-2016 periods. Table 3 shows the estimation results.

Table 3: Firm-level Regressions of Novel Patents by Mega Firms in 1980-2016

	(1)	(2)	(3)	(4)
	# Novel Patents	# Novel Patents	# Novel Patents	# Novel Patents
Mega firm	2.909*** (0.062)	0.185*** (0.037)	0.109*** (0.037)	0.041 (0.050)
Mega firm x Post 2001	0.421*** (0.123)	0.291*** (0.053)	0.347*** (0.053)	0.217*** (0.056)
Lagged IHS(# total patents)		0.896*** (0.005)	0.826*** (0.008)	0.657*** (0.014)
Lagged log sales			-0.023 (0.019)	-0.049* (0.026)
Lagged log employment			0.119*** (0.022)	0.089*** (0.030)
NAICS4 x Year FE	Yes	Yes	Yes	Yes
Firm FE	No	No	No	Yes
Obs.	49,461	38,819	32,879	24,136
Pseudo R-sq	0.45	0.73	0.74	0.76

Notes: The estimates are obtained from Poisson pseudo-maximum likelihood regressions (PPML) with multi-way fixed effects. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. IHS stands for Inverse-Hyperbolic Sine transformation.

Column (1) is the benchmark specification that only controls for industry (four-digit NAICS) by year fixed effects, which, in particular, allows us to check whether the increasing trend of novel patents by mega firms is driven by industry composition.¹⁷ The estimated coefficients indicate that mega firms produced 18.3 times as many novel patents as non-mega firms prior to 2001, while those firms produced 52 percent more novel patents after 2001.¹⁸ In Columns (2) and (3), we examine how much of these differences are explained by observable firm characteristics. Column (2) shows that controlling for the firms' engagement

¹⁷The results are very similar if we do not include these fixed effects.

¹⁸The interpretation of the coefficients is derived from $\exp(2.909) = 18.3$ and $\exp(0.421)-1 = 0.52$.

in overall patenting, measured by the lagged value of total number of patents (applying inverse-hyperbolic sine transformation to accommodate zero patents), reduces the vast majority of the difference in the outcome variable between mega firms and non-mega firms prior to 2001, while a large and significant difference still exists after 2001. In Column (3), we additionally control for firm sales and employment and once again find that mega firms produce relatively more novel patents compared to other firms after 2001, suggesting that the trend toward increasing novel patents by mega firms holds independently of the increase in their total number of patents or size. Finally, in Column (4), we include firm fixed effects to also account for time-invariant unobservable factors. In this specification, the identification is based on within-firm changes in their mega-firm “status” (from non-mega firms to mega firms and vice versa) from year $t - 1$ to year t .¹⁹ Notably, there is no longer any difference in the number of novel patents between mega firms and non-mega firms prior to 2001, but we still find that mega firms produced 24 percent more novel patents than non-mega firms after 2001 even after controlling for firm fixed effects. This suggests that in recent decades, firms are more (less) likely to produce novel patents as they pass the ranking threshold from (to) a non-mega firm to (from) a mega firm.

3.3 Trends in Mega Firms’ Share of Novel and Impactful Innovation: An Alternative Measure

We further examine whether the rise in mega firms’ contributions to novel innovation since 2001 remains evident when using alternative measures of novelty. Specifically, we employ the widely recognized indicator of patent importance developed by Kelly et al. (2021), which leverages natural language processing techniques to establish links between new patents and both existing and subsequent patents. As mentioned in Section 2.2, Kelly et al. identify a patent as important if its content is distinct from prior patents (is novel) but similar to future

¹⁹Over the sample period, there are 233 firm-year observations that switch the mega firm indicator from the previous year.

patents (is impactful). Thus, the indicator of patent importance identifies advancements at the technological frontier that serve as a foundation for subsequent inventions. Since Kelly et al. adopt a fundamentally different measurement approach from ours, applying this metric enables us to assess the robustness of our findings regarding trends in mega firms’ role in creating novel innovation. Additionally, it sheds light on whether mega firms’ contributions to both novel *and* impactful patents have increased over time.²⁰

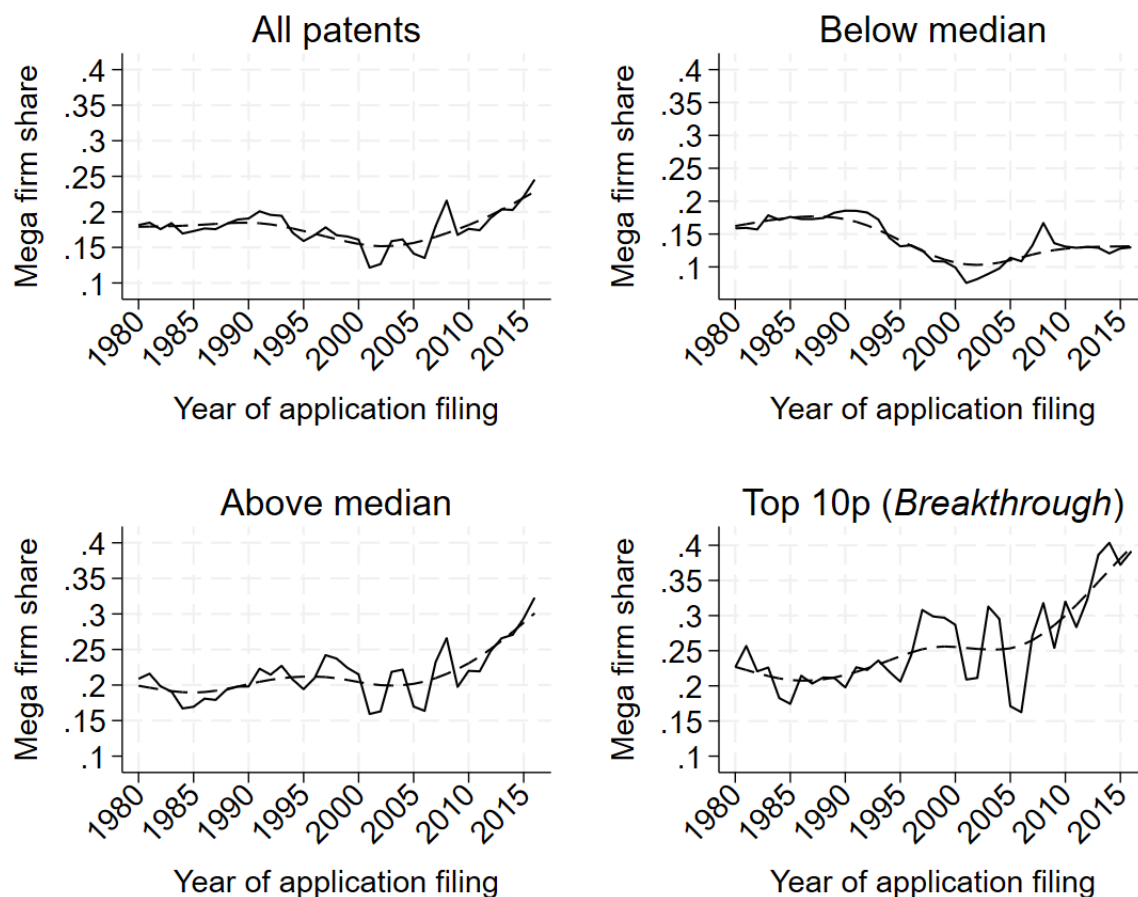
We categorize patents into groups based on their importance, as identified by Kelly et al., ranging from the bottom 10 percent (the least novel and least impactful) to the top 10 percent (the most novel and impactful), with the latter classified as breakthrough patents. Figure 4 presents the share of patent applications filed by mega firms as a proportion of all patent applications (solid line) within each group, alongside its smoothed trend (dashed line), estimated using the HP filter.

We find that the share of patent applications filed by mega firms among the most novel and impactful patents has risen rapidly since the early 2000s. Specifically, the share of breakthrough patents accounted for by mega firms nearly doubled from 21% in 2001 to 39% in 2016, with a large increase observed among patents in the above-median group. By contrast, mega firms’ presence in the less novel and less impactful patents (below median) declined notably in the 1990s and has not rebounded in any meaningful way since. These findings indicate that the growing importance of mega firms extends beyond novel innovation and encompasses both novel and impactful innovation. They also suggest that our main results are not driven by the specific measure of novelty we use.²¹

²⁰We present further evidence on the impact of novel patents produced by mega firms using the follow-on patents measure in the next section.

²¹We also examined the trend in the share of mega firms using alternative text-based novelty only measures from Arts et al. (2021) and Kalyani (2024). The findings using those alternative measures are also similar to the findings using our measure of novel patents, although the degree of similarity varies according to how many “new bigrams” are used to define novel patents.

Figure 4: Share of Patents Filed by Mega Firms Across Novelty and Impact Groups



Source: Authors' own calculation using the USPTO patent data matched with Compustat data.

Notes: The figure displays the share of patent applications filed by mega firms as a proportion of all patent applications from 1980 to 2016 (solid line) within each novelty and impact group, along with its smoothed trend (dashed line) estimated using the Hodrick-Prescott filter with a smoothing parameter of 100. The title in each panel indicates the respective novelty and impact group, based on the measure developed by Kelly et al. (2021). For instance, the top-left panel represents the bottom 10% group (i.e., the least novel and least impactful patents), while the bottom-right panel corresponds to the top 10% group (i.e., the most novel and impactful patents, labeled as breakthrough patents by Kelly et al. (2021)). Mega firms are defined as the top 50 firms by sales in Compustat in each year.

4 Knowledge Diffusion and Market Concentration

Having established the growing contribution of mega firms to novel innovation, we now examine the technological impact and diffusion of knowledge stemming from these innovations, and how these innovations are correlated with mega firms’ subsequent market share.

4.1 Follow-on Patents

While we have already shown that mega firms’ share of breakthrough patents has increased substantially in recent decades based on Kelly et al.’s measure—which assesses a patent’s novelty and its impact using linguistic similarity—we complement this finding with a measure that captures more direct reuse of technological combinations. Specifically, we draw on the approach of Pezzoni et al. (2022), who measure the impact of a novel patent by counting the number of *follow-on* patents—defined as patents that reuse the same new combination of technological components first introduced by the novel patent. This measure captures the extent to which a novel patent opens new technological trajectories that are subsequently built upon by others, based on information from technological classification.²²

To this end, we employ a difference-in-differences style framework to estimate how the number of follow-on patents and related outcomes evolved over time for novel patents produced by mega firms relative to those by non-mega firms. Table 4 presents the results.

²²Forward citations, a widely used measure of a patent’s influence on future innovations, are less appealing in our context because a novel patent may be cited for reasons unrelated to the new combination it introduces. Nevertheless, our main findings in Section 4 remain robust when using citation-based measures. See Online Appendix A.9.

Table 4: Follow-on Patents on Novel Patents by Mega Firms

	(1)	(2)	(3)
	# Follow-on Patents	No Follow-on	Hit
Mega firm	-0.070** (0.027)	0.005 (0.005)	-0.001 (0.001)
Mega firm x Post 2001	0.190*** (0.037)	-0.015** (0.007)	0.005*** (0.002)
Section x Year FE	Yes	Yes	Yes
Obs.	147,318	147,318	147,318
Pseudo R-sq	0.07		
R-sq		0.04	0.00

Notes: Column (1) shows the result from a Poisson pseudo-maximum likelihood regression (PPML) with multi-way fixed effects, where the outcome variable is the number of follow-on patents. Columns (2) and (3) are results from linear probability OLS regressions. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In Column (1), the outcome variable is the number of follow-on patents filed in the five years after the novel patent’s application year. In Column (2), the outcome is an indicator for whether no follow-on patents were filed within this period. Column (1) reports estimates from a Poisson pseudo-maximum likelihood regression, while Column (2) presents results from a linear probability model. The results in Column (1) indicate that, prior to 2001, novel patents by mega firms generated 7% fewer follow-on patents than those by non-mega firms, but after 2001, they produced 13% more.²³ The results in Column (2) suggest that the likelihood of having no follow-on patents within five years was similar for mega and non-mega firms before 2001, but was 1.5 percentage points lower for mega firms thereafter.

To assess whether mega firms produce not only novel patents but also the most successful ones, we follow Pezzoni et al. (2022) and identify “hit” novel patents. In the baseline specification, hits are defined as novel patents that rank in the top one percentile of follow-on patent counts within their main CPC section over the first five years (results are robust to alternative thresholds, such as the top five percentile). Column (3) of Table 4 presents estimates from a linear probability model, where the outcome variable is an indicator for

²³ $\exp(0.190-0.070)-1=0.127$.

whether a novel patent becomes a hit. The results show no significant difference between mega and non-mega firms prior to 2001, but after 2001, novel patents by mega firms were 0.5 percentage points more likely to become hits, compared to a baseline probability of 1.1% for non-mega firms.

4.2 Self-follow-on Patents

Pioneering a new combination is inherently experimental and can be viewed as risk-taking reducing the uncertainty in viability of a new technological space and thus facilitating follow-on innovation. It has been argued that a possible reason for declining business dynamism in the U.S. and elsewhere may be a slowdown in new knowledge diffusion from leading to laggard firms (e.g., Akcigit and Ates, 2021). One rather straightforward way to examine knowledge spillover from leading to laggard firms is by looking at the dissemination of the follow-on patents beyond the focal firm that generated the novel patent—whether follow-on patents are assigned to the same or to different firms. If the share of follow-on patents generated by the same firm that came up with the new combination (self-follow-on rate, hereafter) is increasing over time, it can perhaps be interpreted as evidence of a slowdown in new knowledge diffusion.

Table 5 presents regression results where the dependent variable is the self-follow-on rate within the first five years. Column (1) includes an indicator for mega firms as the key independent variable, while Column (2) adds an interaction term between the post-2001 dummy and the mega firm indicator. Columns (3) and (4) replicate this analysis for “hit” novel patents, defined as before.²⁴

²⁴Novel patents without any follow-on patents are excluded from these regressions since the dependent variable is the share of self-follow-on among all follow-on patents.

Table 5: Self-follow-on Rates of Novel Patents over the First Five Years

	(1)	(2)	(3)	(4)
	Self-follow-on Rate	Self-follow-on Rate	Self-follow-on Rate	Self-follow-on Rate
Mega firm	-0.003 (0.006)	0.004 (0.006)	0.015 (0.030)	0.016 (0.035)
Mega firm x Post 2001	-0.011 (0.008)	-0.002 (0.008)	-0.036 (0.038)	-0.002 (0.042)
Post 2001	0.015*** (0.003)		-0.010 (0.015)	
Section x Year FE	No	Yes	No	Yes
Condition on Hits	No	No	Yes	Yes
Obs.	85,656	85,656	1,763	1,725
R-sq	0.00	0.02	0.00	0.23

Notes: All columns show the result from linear probability OLS regressions, where the outcome variable is the share of follow-on patents that are produced by the firm that initially produced the corresponding novel patent. By construction, the estimation includes only novel patents that have at least one follow-on patent. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The estimation results in Column (1) do provide some evidence of a slowdown in knowledge diffusion for novel patents overall, as indicated by the positive and significant coefficient on the post-2001 dummy variable. However, mega firms exhibit a self-follow-on rate comparable to that of all firms, with no significant change over time. Also, as can be seen from Column (3), there is no evidence of any decline in knowledge diffusion from the most important “hit” novel patents, either for all firms or for the subset of mega firms, before and after 2001. Thus, at this level of analysis, we find no evidence supporting a decline in knowledge diffusion from novel combinations pioneered by mega firms.

4.3 Market Share of Mega Firms

The preceding analyses indicate that novel innovation by mega firms has exerted a greater influence on subsequent innovation since the early 2000s, with no evidence of a slowdown in knowledge diffusion. At first glance, these findings appear to be at odds with Akcigit and Ates (2021), who argue that the rise in market concentration among the largest firms

has been accompanied by their strategic patent accumulation and a slowdown in knowledge diffusion. To better understand this apparent discrepancy, we examine how novel and non-novel innovation by mega firms are correlated with their subsequent industry-level market shares.

Table 6 reports regressions of mega firms’ sales share—defined as mega firms’ sales divided by the total sales of all Compustat firms in the same industry—on the lagged log numbers of novel and non-novel patents filed by mega firms. The specifications include industry and year fixed effects, as well as time-varying industry controls such as the total number of firms and the total number of patent applications filed in the industry. The results reveal a clear difference between novel and non-novel innovation in their association with mega firms’ market share. Column (1) shows that a 1 percent increase in non-novel patent filings is associated with roughly a 10 percent increase in mega firms’ market share. By contrast, the number of novel patents is not significantly correlated with market share and, if anything, the estimated relationship is negative. Column (2) further interacts these variables with a post-2001 indicator. The results suggest that these patterns are present in both the pre- and post-2001 periods.

Taken together with the time trends documented in Section 3, these results point to two distinct modes of innovation by mega firms. On the one hand, non-novel patenting by mega firms has outpaced that of non-mega firms, and this pattern is positively associated with growth in their market shares. This appears to be consistent with the notion that the largest firms do accumulate strategic patent portfolios that can potentially be used to stifle competition. On the other hand, novel patenting by mega firms is not associated with greater market share and instead appears to generate knowledge spillovers to other firms, thereby creating room for their growth. Thus, the increase in novel patents generated by mega firms, which is a new phenomenon in recent decades, appears to point to those firms contributing to technological progress in the U.S. economy overall.

Table 6: Patents by Mega Firms and Their Market Share

	(1)	(2)
	Mega Firms' Sales Share	Mega Firms' Sales Share
Novel Patents by Mega Firms	-0.0490* (0.0257)	-0.0142 (0.0280)
Novel Patents by Mega Firms x Post 2001		-0.0596* (0.0332)
Non-novel Patents by Mega Firms	0.0949*** (0.0157)	0.0796*** (0.0181)
Non-novel Patents by Mega Firms x Post 2001		0.0241 (0.0195)
Year FE	Yes	Yes
Industry FE	Yes	Yes
Controls	Yes	Yes
Observations	6,074	6,074
R-squared	0.8025	0.8044

Notes: These OLS regressions are estimated using an industry-year panel, with industries defined at the four-digit NAICS level. Mega Firms' Sales Share is measured as mega firms' sales divided by the total sales of all Compustat firms in the same industry. Novel Patents by Mega Firms and Non-novel Patents by Mega Firms denote the one-year-lagged log numbers of novel and non-novel patents, respectively, filed by mega firms in the industry. We add one to each patent count to accommodate zeros when taking logs. Controls include the total number of firms and the total number of patent filings at the industry-year level. Standard errors are clustered at the industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5 Potential Driving Factors

The coincidence of declining trends in the efficiency of R&D investment and business dynamism, together with the rise of “superstar” firms, has led economists to consider whether these developments are related. Our empirical analysis of recent trends in novel patenting, however, suggests that at least the very largest firms—mega firms—have contributed more, not less, to the production of novel patents relative to other firms, with the turnaround occurring in the early 2000s. A key question, then, is what factors may be associated with this increase in novel innovation by mega firms. Although causally identifying the underlying drivers is beyond the scope of this paper, we examine two contemporaneous trends that may help shed light on this question.

5.1 Cash Holdings by Mega Firms

A large body of research in corporate finance has documented that R&D is difficult to finance externally due to its low pledgeability, uncertain outcomes, and reliance on trial-and-error processes (for survey, see Kerr and Nanda, 2015). As a result, internal sources—particularly cash holdings—play a key role in funding R&D investment (Brown et al., 2009; Hall and Lerner, 2010; He and Wintoki, 2016). Such financial constraints are likely to be especially salient for projects aimed at generating novel technological combinations, which inherently involve even higher uncertainty. Building on these insights, we investigate whether the dynamics of cash holdings by mega firms can help explain the trend shown in Figure 1.

We begin by examining the relationship between cash holdings and novel innovation. Krieger et al. (2022) provide a framework well-suited to this context. They develop a model of costly external financing and risky R&D in which financial frictions induce firms to behave as if they are risk averse, even where shareholders are risk neutral. This endogenous risk aversion arises because the marginal cost of external funds rises sharply once internal liquidity is depleted, rendering the value function locally concave. As a result, larger cash holdings reduce the likelihood of costly financing shortfalls and increase the firm’s willingness to undertake risky R&D investments. Consistent with this mechanism, Krieger et al. provide empirical evidence in the context of drug development that higher cash flows lead firms to invest more in R&D, particularly in riskier, novel projects.

We test whether the predictions of Krieger et al. carry over to our setting. To this end, we examine the dynamic conditional correlation between changes in cash holdings and subsequent innovation outcomes—both total and novel—using local projections. Specifically, we estimate the following equation:

$$\Delta_{t+k,t-1}^{\%} Y_i = \alpha + \beta_k \Delta_{t,t-1}^{\%} cash_i + \gamma' X_{i,t} + \theta_i + \eta_t + \epsilon_{t+k}, \quad (4)$$

where i indexes firm and t denotes years. $\Delta_{t+k,t-1}^{\%} Y_i$ represents the percent change in the

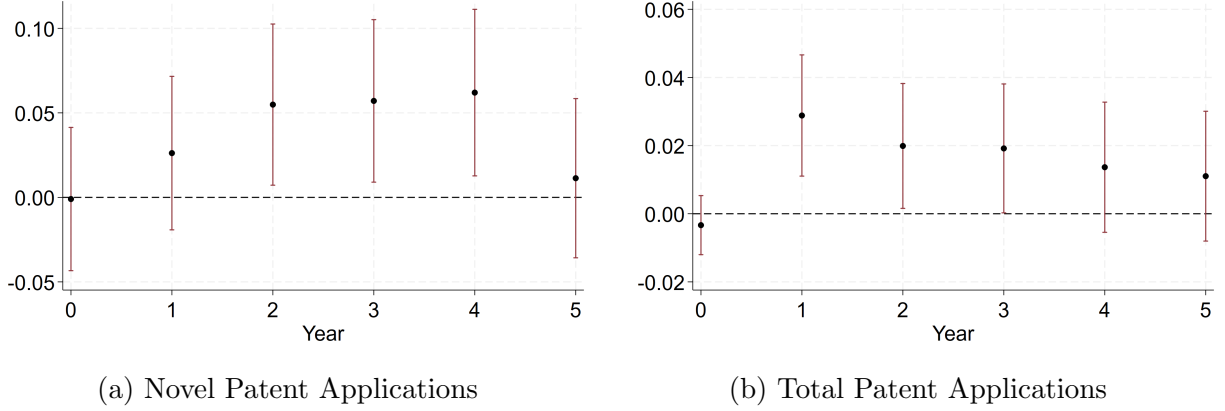
number of patent applications between $t - 1$ and $t + k$, and $\Delta_{t,t-1}^{\%} \text{cash}_i$ is the percent change in cash holdings between $t - 1$ and t .²⁵ The vector $X_{i,t}$ includes control variables such as log assets and total patent applications, each measured at t and with three lags (through $t - 3$), as well as cash holdings with three lags ($t - 1$ to $t - 3$), while the results are robust to alternative lag specifications. Firm and year fixed effects are denoted as θ_i and η_t , respectively. We estimate this equation for six years, starting from year t , that is, for $k = 0, 1, 2, 3, 4, 5$. The coefficient of interest is β_k , which indicates the percent change in novel patent applications between year $t - 1$ and $t + k$ following a one percent increase in cash holdings in year t holding everything else constant.

Figure 5 displays the estimated β for $k = 0, 1, 2, 3, 4$, and 5 . The left and right panels display results where Y_{it} represents novel patent applications and total patent applications, respectively. The left panel shows that following a one percent increase in cash holdings in year t , the number of novel patent applications gradually rises, reaching a 6.2% increase by year $t+4$. In contrast, the right panel indicates a more immediate increase in total patent applications, though the peak effect is small (2.9%) compared to novel patent applications. Overall, consistent with the findings of Krieger et al. (2022), increases in cash holdings are associated with greater subsequent patenting activity, with more than twice the magnitude for novel patents compared to all patents.

The relationship above, although it is not causally identified, motivates us to next examine the evolution of differences in cash holdings between mega firms and non-mega firms. To this end, we estimate a difference-in-differences style regression using our Compustat firm

²⁵We define percent change of Y between $t - 1$ and t as $\frac{Y_t - Y_{t-1}}{(Y_t + Y_{t-1})/2}$ and similarly for cash holdings. This is a simple relative change measure developed by Tornqvist et al. (1985) and popularized by Davis et al. (1996). This measure is identical to the log difference under a second-order approximation and offers several advantages. First, it is symmetric for increases and decreases, similar to the log-difference measure, but unlike the log difference, it accommodates zeros. This property is particularly relevant to this analysis, as many firm-year observations in our dataset have zero novel patent applications. Second, this measure aligns with the recommendations of Chen and Roth (2024) and Mullahy and Norton (2024) who advocate for scale-independent change measures. In contrast, transformations such as $\log(1 + x)$ or inverse hyperbolic sine are scale-dependent and therefore less suitable for outcome measures. Lastly, this measure provides a better model fit when dealing with highly skewed variables compared to the traditional percent change measure.

Figure 5: Percent Change in Patent Applications Following An Increase In Cash Holdings



Notes: The figures show estimated values of β_j in Equation (4), where the left panel and right panel use the percent change in novel patent applications and total patent applications, respectively. Vertical bands represent 95% confidence intervals. Standard errors are clustered at the firm level.

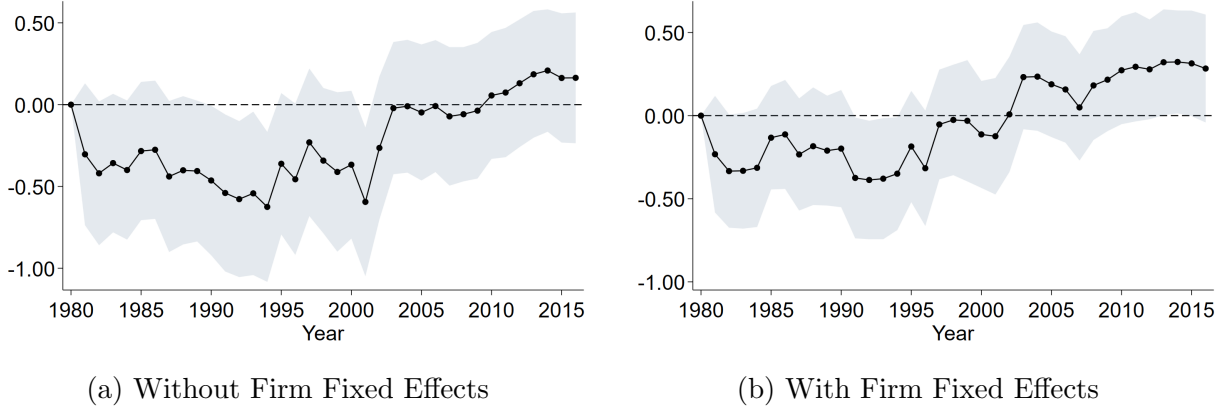
panel data:

$$Y_{it} = \alpha + \beta \cdot Mega_{it} + \sum_{\tau=1981}^{2016} \eta_{\tau} \cdot \mathcal{I}_{\tau=t} + \sum_{\tau=1981}^{2016} \delta_{\tau} \cdot \mathcal{I}_{\tau=t} \cdot Mega_{it} + X'_{it}\gamma + \theta_i + \epsilon_{it}, \quad (5)$$

where Y_{it} represents the log of cash holdings for firm i in year t , and $Mega_{it}$ is an indicator for mega firms. $\mathcal{I}_{\tau=t}$ are year dummies and the term η_{τ} denotes year fixed effects, while δ_{τ} is the coefficient of interest, tracing the evolution of differences in cash holdings between mega firms and non-mega firms. The control variables X_{it} include log assets as the baseline control, aligning with the literature that measures cash holdings using the cash-to-asset ratios. Results are robust to additionally controlling for employment or sales.

Figure 6 presents the estimated δ_{τ} using 1980 as the base year. The left panel displays the results without firm fixed effects (θ_i), while the right panel includes them. We find that cash holdings of mega firms, relative to non-mega firms, declined from 1980 to the mid-1990s but have steadily increased since then. The sharp rebound since the early 2000s is evident regardless of whether we isolate within-firm changes in cash holdings (i.e., firms holding more cash as they become mega firms) or also account for compositional shifts (i.e., firms

Figure 6: Log Difference in Cash Holdings Between Mega Firms and Non-mega Firms



Notes: The figure shows the estimated difference in log cash holdings between mega firms and non-mega firms from 1980 until 2016 using 1980 as the base year, that is, δ_τ in Equation (5). The shaded area displays 95% confidence intervals around the estimated coefficients, constructed using heteroskedasticity robust standard errors.

with more cash becoming mega firms). It is interesting to note that the time pattern of the evolution of cash holdings by mega firms relative to non-mega firms in Figure 6 follows the same U-shaped pattern—rather than a monotonic increase—as the trend observed in Figure 1. Combined with the findings that increased cash holdings were associated with more novel patents in Figure 5, this supports the notion that increased cash holdings by mega firms may have facilitated their increased contribution to novel innovation.²⁶

5.2 Technological Scope of Inventor Teams

Recall that, by definition, novel innovation involves combining technological components that have not previously been used together. This concept is closely aligned with recombinant innovation as described by Weitzman (1998), who shows that the probability of generating new combinations increases with the breadth of the existing knowledge base. Also, as the stock of knowledge accumulates, achieving novel breakthroughs often requires not only larger

²⁶Figure 5 shows that the positive association peaks four years after the initial increase in cash holdings. Since the increase in relative cash holdings by mega firms occurs in the early 2000s, this dynamic relationship is consistent with the rise in mega firms' novel patent applications around the mid-2000s shown in Figure 2.

but also diverse teams of inventors with increasingly specialized expertise (Jones, 2009; Freund, 2025). We therefore conjecture that the pace of novel innovation is affected by the breadth (scope) of the technological expertise embodied in inventor teams that firms employ.

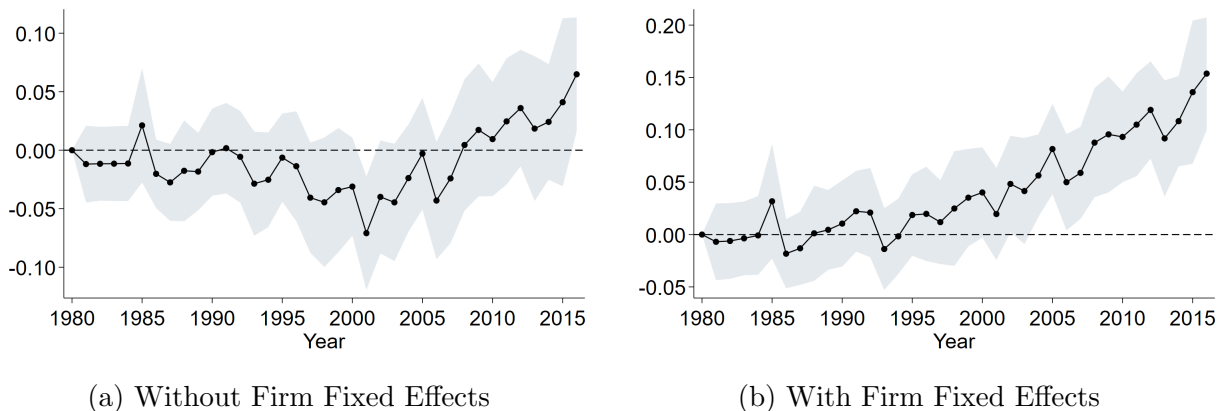
Based on this idea, we replace the dependent variable Y_{it} with the average number of technological areas in which inventor teams have expertise when estimating Equation (5). Since technological scope may mechanically increase with increases in team size, we control for firms’ average inventor team size, though results remain robust to excluding this control.²⁷ To construct the variable capturing an inventor’s technological expertise in year t , we identify the CPC technology group in which the inventor has filed the largest number of patents over their entire cumulative patenting history up to year t . We then count the number of distinct technological areas across all members of the inventor team associated with patent applications by firm i in year t .

Figure 7 presents the evolution of differences in the technological scope of inventor teams between mega and non-mega firms. The left panel, once again, presents results without firm fixed effects, capturing both within-firm changes and compositional shifts. The relative technological scope of inventor teams at mega firms follows a U-shaped pattern, with a trough around 2000. The right panel incorporates firm fixed effects, isolating within-firm variations. From the mid-1990s, firms began broadening the technological scope of their inventor teams as they became mega firms. Thus, comparing the two panels, we can see that the decline in technological scope in the last two decades of the 20th century was driven by compositional shifts—an increasing share of firms with narrower technological scope among mega firms, while after 2000, mega firms with a broader technological scope took the center stage.

The estimation results suggest that the competitive advantage of mega firms in producing novel innovations may be accruing to those of them that assembled not just larger but more

²⁷The literature has noted the rising concentration of inventors in the largest firms (Akcigit and Goldschlag, 2023), but we observe this trend persistently throughout all four decades in our sample and therefore it cannot, by itself, account for the decline in novel patenting by mega firms during the 1980s and 1990s, followed by a sharp reversal after 2000.

Figure 7: Log Difference in Technology Scope Between Mega Firms and Non-mega Firms



Notes: The figures present estimated values of δ_τ in Equation (5) from 1980 until 2016, where the outcome variable is the log of the average number of CPC technological groups in which inventor teams have expertise at the firm-year level. 1980 is used as the base year. The left panel excludes firm fixed effects, while the right panel includes them. The shaded area represents 95% confidence intervals, and robust standard errors are used.

diverse, in terms of their technological expertise, inventor teams. The findings are also robust to including additional firm size measures, such as employment and sales. Overall, these results support the hypothesis that an increasing scope of technological expertise at mega firms has facilitated the rapid reversal in their contribution to novel innovations since the early 2000s.

5.3 Discussion

It is useful to interpret the findings above in light of recent literature on the rise of mega firms, increase in markups, and decline in business dynamism. A growing body of research highlights the rising use of intangible inputs—such as information technology and software—as a key driver of these trends. For example, De Ridder (2024) emphasizes intangible inputs that are embedded in physical products or services, such as a phone’s operating system or a car’s drive-by-wire system. These technologies typically require substantial upfront investment in development but are nearly costless to replicate once deployed, allowing firms

to scale at low marginal cost. As a result, firms that can successfully develop and deploy such technologies are able to grow faster and dominate their markets, contributing to rising concentration. Other studies highlight how Information and Communication Technologies (ICT), when integrated into firms’ production processes, improve efficiency and reduce overhead costs, facilitating expansion into new geographic and product markets (Aghion et al., 2023; Hsieh and Rossi-Hansberg, 2023). While the specific mechanisms differ, these studies consistently show that economic activity is increasingly concentrated in firms that are better able to deploy intangible inputs than others, leading to higher market concentration.

If it is true that a substantial subset of novel innovation is linked to the development and deployment of intangible inputs, mega firms are increasingly securing advantageous positions in producing such innovation for the reasons discussed in Sections 5.1 and 5.2 above.²⁸ The rise in their novel combinations may thus be seen as a reflection of the same intangible capabilities that have enabled mega firms to expand their market shares, particularly since the early 2000s. However, as shown in Section 4.3, it is non-novel patenting—rather than novel innovation itself—that is directly associated with mega firms’ market share growth. This suggests that even if the underlying capacity to develop and deploy intangible inputs drives both novel innovation and market dominance, novel patents contribute to technological progress more broadly rather than serving as a direct vehicle for market share expansion. Importantly, as shown in Section 4, these innovations generate meaningful knowledge spillovers beyond firm boundaries—a relatively overlooked but potentially beneficial aspect of mega firms’ growing role.

In addition, the findings in Section 5.2 offer a new perspective on the reallocation of

²⁸Indeed, there are numerous examples of novel innovation in which firms combine (intangible) ICT with non-ICT components in their products, services or production processes. For example, Amazon has filed several novel patents that combine technologies in Wireless Communication Networks (CPC code H04W) with Transport or Storage Devices (B65G) to improve inventory management through robots that receive instructions from a central computer system. Amazon has also filed patents that integrate components from Equipment for Fitting in or to Aircraft (B64D) with Data Processing Systems or Methods (G06Q) to enable drone-based package delivery. Alphabet (Google) has likewise filed numerous novel patents that combine technologies classified under Vehicles (B60) with those under Computing (G06), aimed at developing autonomous vehicles—commercialized through its spin-off subsidiary, Waymo.

inventors toward large incumbents documented in Akcigit and Goldschlag (2023). While they show that inventors have increasingly moved to large firms since 2000, with declines in individual productivity—measured by the number of patent applications and citations—our results highlight how mega firms are organizing diverse inventor teams and actively leveraging their expertise. In a related vein, Jin et al. (2023) analyze M&A activity by the five leading technology platforms—Alphabet (Google), Apple, Facebook, Amazon, and Microsoft (GAFAM)—and find that GAFAM’s acquisitions span a broader range of technology categories than those of other major acquirers, such as top private equity firms. This evidence suggests that mega firms may be expanding their technological scope and acquiring inventors through firm acquisitions, although more systematic analysis is needed.

6 Conclusion

Our empirical analysis reveals a U-shaped trend in the share of novel patents produced by mega firms, declining from 1980 to the early 2000s before rebounding sharply to reach its highest level by 2016. This resurgence reflects not only an increase in mega firms’ overall patenting activity but also a heightened propensity to engage in novel innovation, as further evidenced by firm-level analysis. Since 2001, novel patents by mega firms have demonstrated greater technological impact, generating more follow-on patents and being more likely to become “hits” compared to those by non-mega firms. We also find that the share of novel patents in U.S. patent applications overall had declined over several decades until the mid-2000s, followed by a rebound since then, starting several years after the start of the rebound in the share of novel patents by mega firms. Together, these findings suggest that mega firms might be increasingly shaping new technological trajectories in the U.S. economy.

Our findings on knowledge diffusion indicate that the self-follow-on rate—the proportion of follow-on patents generated by the same firm—has remained stable for mega firms over time, showing no evidence of impeding knowledge spillover specifically for their novel patents.

This stability, combined with the higher volume of follow-on patents post-2001 for these novel innovations, suggests that mega firms are facilitating knowledge diffusion to other firms through this selected subset of innovations. While mega firms may restrict knowledge diffusion tied to other types of patents, our analysis highlights that this pattern does not extend to their novel innovations. We also explore potential drivers of this trend, finding suggestive evidence that the relative increase in cash holdings since the early 2000s may have enabled mega firms to fund risky, experimental R&D, while the broadening technological scope of their inventor teams may have enhanced their capacity to pioneer new combinations.

The share of economic activities accounted for by mega firms has dramatically increased over the past several decades, and their innovation behavior has profound implications for economic growth, technological progress, and the appropriate policy response. While mega firms may be strategically protecting their technological superiority in certain dimensions, our evidence suggests that, especially in recent years, they are also leading technological experimentation through novel innovations that enable follow-on innovation by others. If mega firms were predominantly hindering knowledge spillover, there might be a case for considering regulatory intervention. However, given their role as key actors in generating novel technologies, the impact of such measures requires careful consideration to avoid unintended consequences. Understanding the balance between these countervailing forces remains a critical research agenda for informing policy in this debate.

Acknowledgements

We thank Sina Ates, Pierre Azoulay, Vojislav Maksimovic, and participants at the Workshop on Entrepreneurial Finance and Innovation, Technology & Policy Research Initiative Seminar, 2023 NBER Megafirms and the Economy conference, Ed Snider Center at the University of Maryland, Hitotsubashi University, Federal Reserve Board, and 2025 BSE Summer Forum for helpful comments and suggestions.

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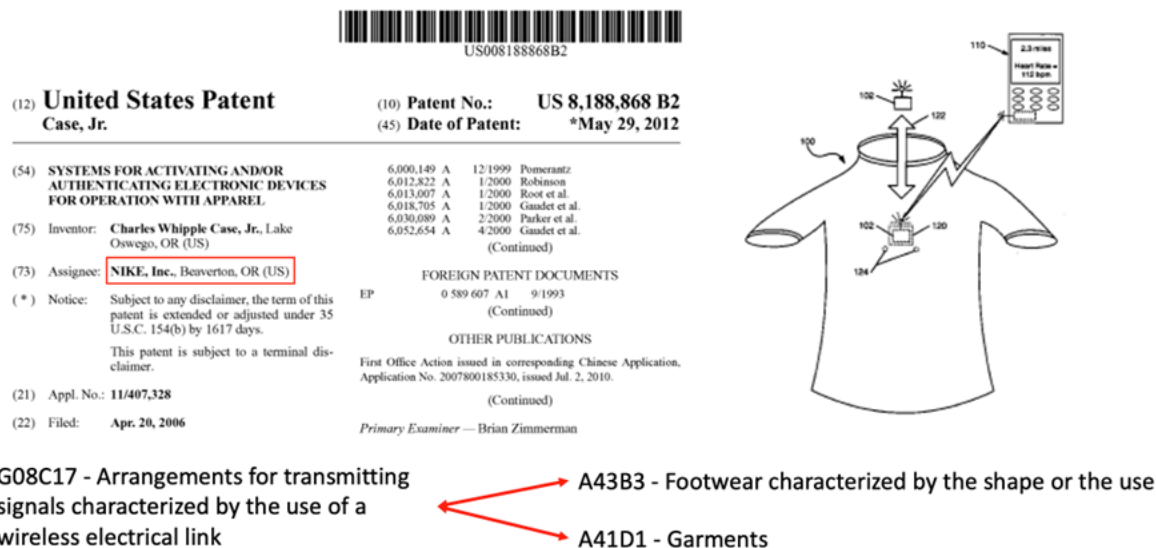
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A Online Appendix to Mega Firms and New Technological Trajectories in the U.S.

A.1 Examples of Novel Patents

Figure A1: Visual Image of the Novel Patent Example in the Introduction



Source: USPTO.

A.2 USPTO-Compustat Matching

For the analyses involving mega firms, we use S&P's Compustat data to track publicly listed firms in the U.S. We created our own bridge between the U.S. patenting firms in the USPTO patent database and Compustat firms through a standard name-matching and internet-based matching algorithm as in Autor et al. (2020a).

First, we standardize firm names in both datasets using the algorithm provided by the NBER PDP and use the standardized names in the matching process. We define the patenting firms as patent assignees that are located in the U.S. with an assignee type equal to 2 (U.S. company or corporation) in the USPTO data.

The first match procedure involves identifying firms with precisely the same standardized

names in both datasets. Following the previous studies, we do not use address information in Compustat throughout the entire match process as the data only reports information for headquarters, which can be different from the exact address of the establishments that filed patent applications to the USPTO. For the unmatched USPTO firms, we use stem names (standardized firm names without suffixes) to find matches.

For the rest of the unmatched U.S. patenting firms after the standard name matching, we apply an internet-based matching algorithm to identify the same firms in Compustat. Specifically, we put every patent assignee and Compustat firm name into the Google.com search engine, collect the URLs of the top five search results, and identify any given pair of the patent assignee and Compustat firm as the same firm if they share at least two identical search results. If any of these patenting firms remain unmatched, we utilize web-URL information in Compustat and find the corresponding firms if the top five search results of the unmatched patenting firms exactly match the web-URL of the Compustat firms.

For all the remaining unmatched U.S. patenting firms in the USPTO data after the previous steps, we use the NBER PDP and find matches in Compustat. The NBER PDP did extensive manual matching to identify the same firms across the two datasets. Thus, this procedure helps us to reduce the burden of manually searching the unmatched USPTO firms. Lastly, we do our own manual matching to identify matches between the USPTO and Compustat firms. We manually inspect the match results to screen out false matches, especially for firms with many patent applications at the end of each procedure. More details about the matching algorithm are provided in Ding et al. (2022).

The above procedure matches 68.1% of utility patent applications (granted until Sep. 30, 2023) filed by U.S. patenting firms, and 25.6% of U.S. patenting firms to Compustat firms from 1976 to 2016.

A.3 Novel Patent Indicator and Alternative Measures of Novelty

This section presents regression results where the dependent variables capture various alternative measures of patent novelty and creative destruction, with the novel patent indicator as the main independent variable. The tables below report results under different specifications, varying the inclusion of CPC group fixed effects and year fixed effects. The findings remain robust across all fixed-effect combinations.

Table A1: Novel Patents & Number of New Bigrams

	(1) # New Bigrams	(2) # New Bigrams	(3) # New Bigrams	(4) # New Bigrams	(5) # New Bigrams
Novel Patent Indicator	0.777*** (0.019)	0.852*** (0.019)	0.559*** (0.019)	0.769*** (0.019)	0.715*** (0.019)
CPC Group FE	No	Yes	No	Yes	No
Year FE	No	No	Yes	Yes	No
CPC Group x Year FE	No	No	No	No	Yes
Obs.	2,633,463	2,630,141	2,633,463	2,630,141	2,629,129
R-sq	0.00	0.04	0.03	0.06	0.08

Notes: The estimates are obtained from OLS regressions with multi-way fixed effects. “# New Bigrams” indicates the number of new bigrams in patent text, obtained from Arts et al. (2021). Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A2: Novel Patents and Patent Creativity

	(1) Patent Creativity	(2) Patent Creativity	(3) Patent Creativity	(4) Patent Creativity	(5) Patent Creativity
Novel Patent Indicator	0.022*** (0.000)	0.022*** (0.000)	0.018*** (0.000)	0.020*** (0.000)	0.019*** (0.000)
CPC Group FE	No	Yes	No	Yes	No
Year FE	No	No	Yes	Yes	No
CPC Group x Year FE	No	No	No	No	Yes
Obs.	2,578,919	2,575,446	2,578,919	2,575,446	2,574,407
R-sq	0.00	0.06	0.06	0.12	0.16

Notes: The estimates are obtained from OLS regressions with multi-way fixed effects. “Patent Creativity” is the share of new technical bigrams in patents, obtained from Kalyani (2024). Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A3: Novel Patents and Backward Self-citation Ratio

	(1) Backward Self-citation Ratio	(2) Backward Self-citation Ratio	(3) Backward Self-citation Ratio	(4) Backward Self-citation Ratio	(5) Backward Self-citation Ratio
Novel Patent Indicator	-0.031*** (0.001)	-0.026*** (0.001)	-0.032*** (0.001)	-0.027*** (0.001)	-0.026*** (0.001)
CPC Group FE	No	Yes	No	Yes	No
Year FE	No	No	Yes	Yes	No
CPC Group x Year FE	No	No	No	No	Yes
Obs.	2,661,756	2,657,952	2,661,756	2,657,952	2,656,902
R-sq	0.00	0.06	0.00	0.06	0.08

Notes: The estimates are obtained from OLS regressions with multi-way fixed effects. Backward self-citation ratio is defined as the number of citations to a firm's own previous patents divided by the total number of citations. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.4 Novel Patent Indicator and Measures of Impact

This section presents the results of regressions in which the dependent variables represent various measures of patent impact, while the key independent variable is the novel patent indicator. Table A4 illustrates the relationship between stock market valuation and the novel patent indicator.

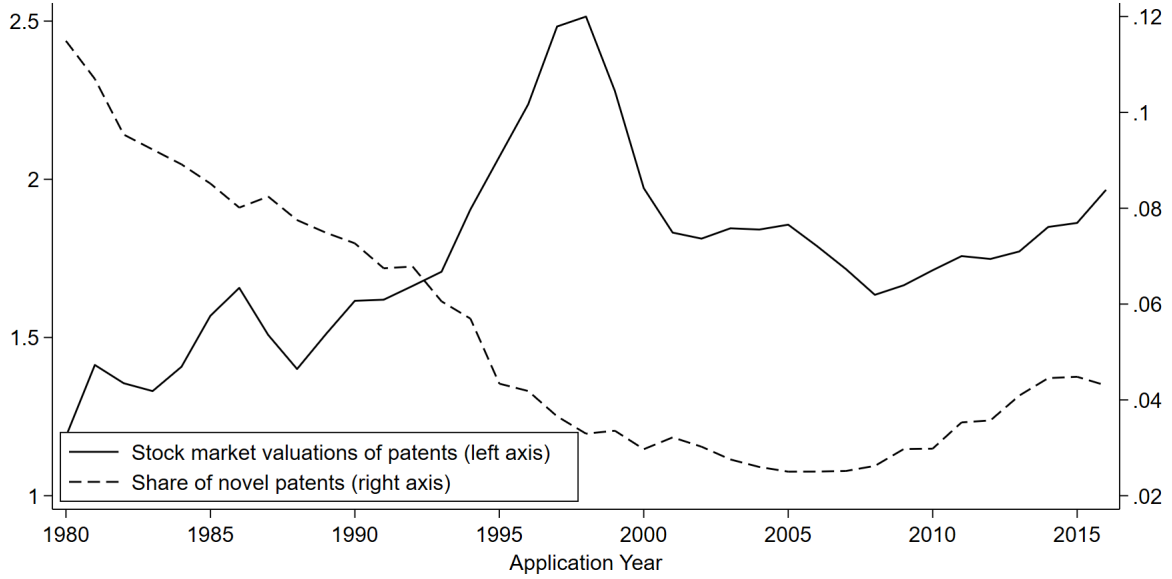
Table A4: Novel Patents and Market Valuation

	(1) Market Valuation	(2) Market Valuation	(3) Market Valuation	(4) Market Valuation	(5) Market Valuation
Novel Patent Indicator	-0.025*** (0.005)	-0.025*** (0.006)	0.043*** (0.005)	0.021*** (0.005)	0.006 (0.005)
CPC Group FE	No	Yes	No	Yes	No
Year FE	No	No	Yes	Yes	No
CPC Group x Year FE	No	No	No	No	Yes
Obs.	1,506,012	1,504,550	1,506,012	1,504,550	1,502,624
R-sq	0.00	0.05	0.04	0.09	0.14

Notes: The estimates are obtained from OLS regressions with multi-way fixed effects. Market valuation is the stock market value of a patent obtained from Kogan et al. (2017). Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

We find that the relationship is negative when year fixed effects are excluded, as shown

Figure A2: Trend in the Average Stock Market Valuations and Share of Novel Patents



Source: Authors' own calculation using the market valuation data from Kogan et al. (2017) and USPTO patent database.
Notes: Stock market valuations are the log of valuations deflated to 1982 (million) dollars using the CPI.

in the first two columns. To understand this result, we plot the time series of the average stock market value of patents (solid line) and the share of novel patents (dashed line) in Figure A2. The two series exhibit opposite trends until around 2008, explaining the negative coefficient when time fixed effects are omitted. The share of novel patents has been declining during this period, a trend that is also broadly consistent with the findings of Arts et al. (2021) and Kalyani (2024). Meanwhile, average stock prices have been rising, largely due to factors unrelated to patent novelty, such as the decline in long-term interest rates and the equity risk premium. Therefore, we deem it more appropriate to control for time fixed effects to better isolate the relationship between the two variables. We also find that the estimated relationship is positive but becomes very small and statistically insignificant when controlling for CPC group-by-year fixed effects. This result suggests that the impact of novel patents on firms' profits may be limited and weak.

Tables A5 and A6 present the relationship between the novel patent indicator and two outcomes: the five-year forward citation count and the breakthrough patent indicator developed by Kelly et al. (2021), respectively. In these cases, controlling for CPC group fixed

effects is crucial for uncovering the positive relationship.

Table A5: Novel Patents and 5-year Forward Citations

	(1) 5-year Forward Citation	(2) 5-year Forward Citation	(3) 5-year Forward Citation	(4) 5-year Forward Citation	(5) 5-year Forward Citation
Novel Patent Indicator	-0.675*** (0.041)	0.528*** (0.042)	-0.428*** (0.041)	0.690*** (0.041)	0.575*** (0.042)
CPC Group FE	No	Yes	No	Yes	No
Year FE	No	No	Yes	Yes	No
CPC Group x Year FE	No	No	No	No	Yes
Obs.	2,842,894	2,838,910	2,842,894	2,838,910	2,837,936
R-sq	0.00	0.03	0.01	0.04	0.07

Notes: The estimates are obtained from OLS regressions with multi-way fixed effects. 5-year forward citation is the number of forward citations received by the patent within the subsequent five years. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6: Novel Patents and Breakthrough Patents

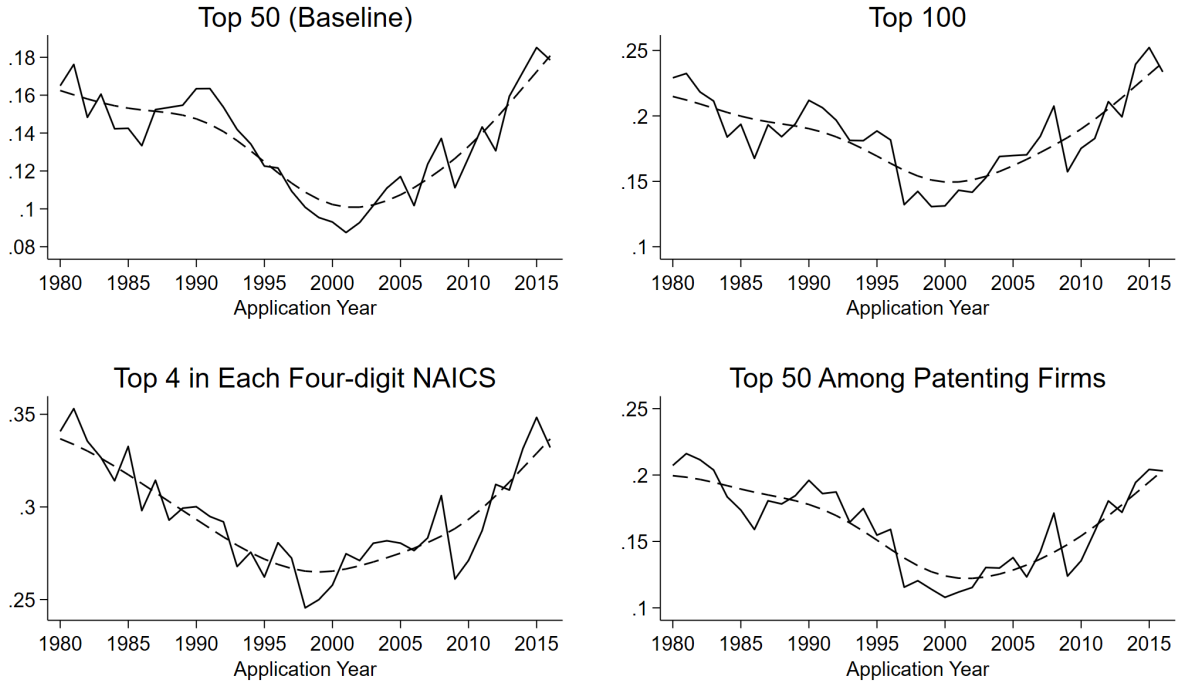
	(1) Breakthrough	(2) Breakthrough	(3) Breakthrough	(4) Breakthrough	(5) Breakthrough
Novel Patent Indicator	-0.039*** (0.001)	0.015*** (0.001)	-0.039*** (0.001)	0.015*** (0.001)	0.004*** (0.001)
CPC Group FE	No	Yes	No	Yes	No
Year FE	No	No	Yes	Yes	No
CPC Group x Year FE	No	No	No	No	Yes
Obs.	2,836,794	2,836,755	2,836,794	2,836,755	2,835,781
R-sq	0.00	0.11	0.02	0.16	0.29

Notes: The estimates are obtained from OLS regressions with multi-way fixed effects. Breakthrough is an indicator of whether a patent is a breakthrough patent, obtained from Kelly et al. (2021). Note that this measure is already residualized with respect to year fixed-effects, so including year fixed effects does not change the coefficients. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.5 Trends in Mega Firms' Share of Novel Patents Under Alternative Definitions of Mega Firms

Figure A3 illustrates the share of novel patent applications filed by mega firms under various definitions. The top-left panel reproduces Figure 1 as a benchmark. The top-right panel

Figure A3: Trends in Mega Firms' Share in Novel Patents Under Alternative Definitions of Mega Firms

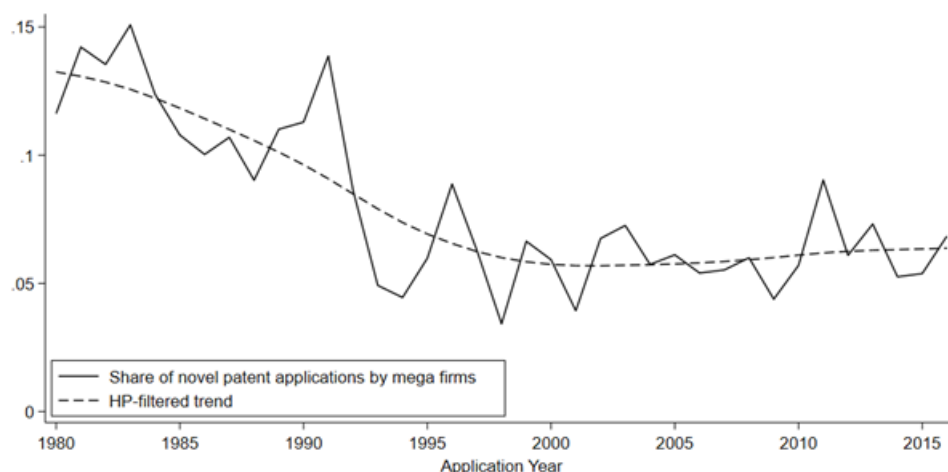


Source: Authors' own calculation using USPTO patent database and COMPUSTAT.

expands the definition of mega firms to the top 100 firms by sales in each year. The bottom-left panel defines mega firms as the top 4 firms by sales in each four-digit NAICS industry. The bottom-right panel restricts mega firms to the top 50 firms by sales in each year, but only among COMPUSTAT firms that file at least one patent, accounting for the fact that patenting firms represent a highly selected subset. We find that the U-shaped trend reported in Figure 1 remains evident across all definitions.

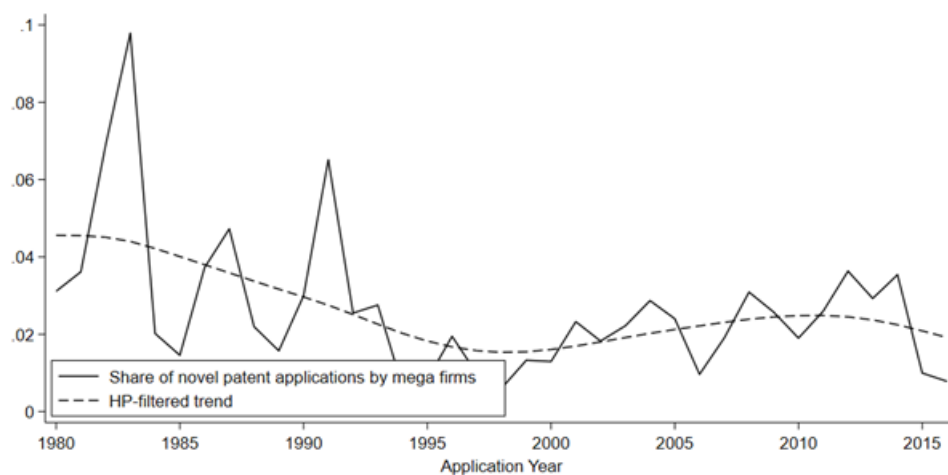
Figure A4 plots mega firms' share of novel patents in the biochemistry and pharmaceutical technology groups. The figure shows that their contribution declined through the late 1990s and never recovered thereafter. Figure A5 illustrates that this trend remains robust even when we redefine mega firms as the top 10 firms in sales in the pharmaceutical industry (NAICS 3254). Therefore, the declining trend in Figure A4 is not driven by the decreasing share of pharmaceutical companies among mega firms.

Figure A4: Trends in Mega Firms' Share in Novel Patents in Biochemistry and Pharmaceutical Technology



Source: Authors' own calculation using USPTO patent database and COMPUSTAT.

Figure A5: Trends in Mega Firms' Share in Novel Patents in Biochemistry and Pharmaceutical Technology (Alternative Definition of Mega Firms)

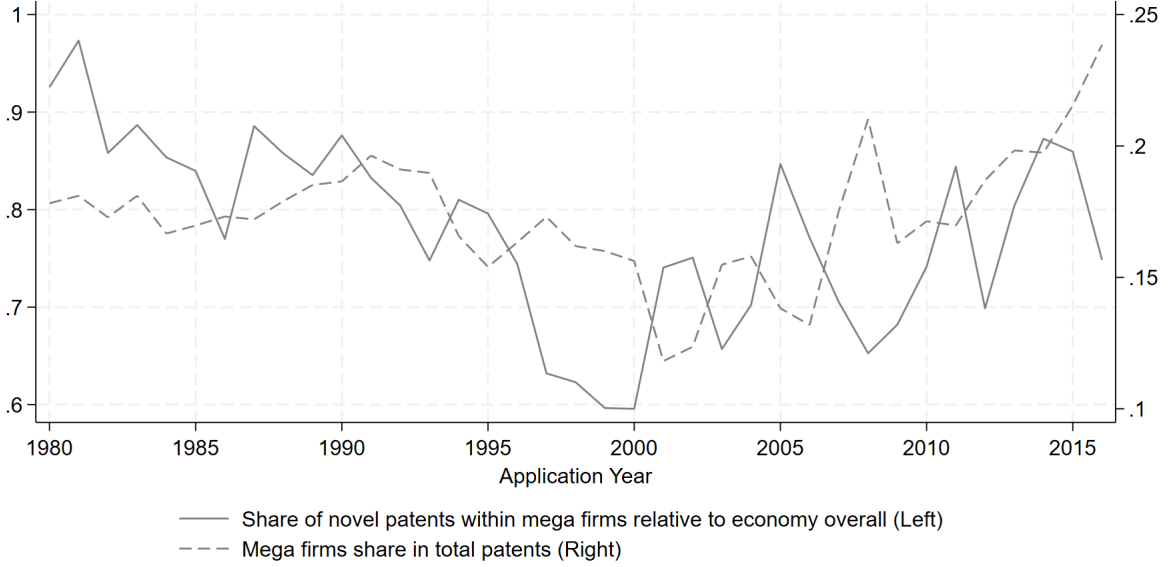


Source: Authors' own calculation using USPTO patent database and COMPUSTAT.

A.6 Decomposition of Mega Firms' Share in Novel Patents

Figure A6 below displays the decomposition of novel patent applications filed by mega firms into (i) the share of novel patents within mega firms relative to that share among all firms (solid line) and (ii) the share of total patents accounted for by mega firms (dashed line). Both time series exhibit a U-shaped pattern with the reversal occurring around 2000.

Figure A6: Decomposition of Mega Firms' Share in Novel Patents



Source: Authors' own calculation using USPTO patent database and COMPUSTAT.

A.7 Trends in Mega Firms' Share of Novel Patents for ICT and non-ICT Patents

Figure A7 shows the share of novel patent applications filed by mega firms for non-ICT patents (left) and for ICT patents (right).

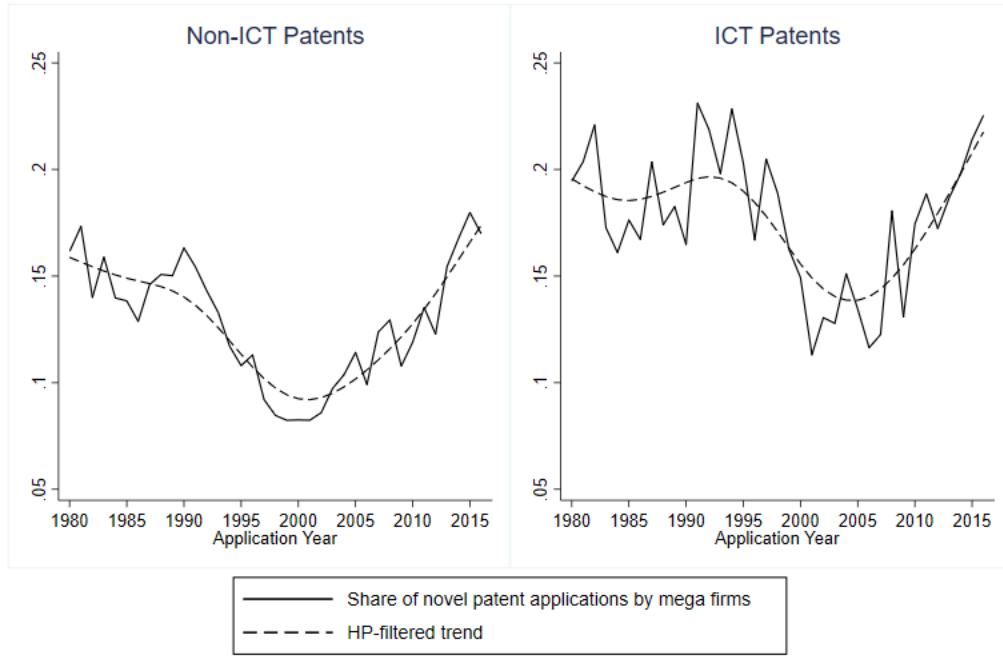
A.8 Contribution by Continuers vs. Change in Composition of Mega Firms

Let $N_{m,t}$ denote the number of novel patent applications filed by mega firms in year t . We decompose the annual change in this quantity, $\Delta N_{m,t} = N_{m,t} - N_{m,t-1}$, as follows:

$$\Delta N_{m,t} = \sum_{i \in S_t^{\text{cont}}} \Delta n_{i,t} + \sum_{i \in S_t^{\text{new}}} (n_{i,t} - 0) + \sum_{i \in S_t^{\text{exit}}} (0 - n_{i,t-1}),$$

where $n_{i,t}$ is the number of novel patent applications by firm i in year t ; S_t^{cont} denotes the set of firms that are mega firms in both years $t-1$ and t (continuers); S_t^{new} refers to firms

Figure A7: Trends in Mega Firms' Share in Novel Patents for ICT and non-ICT Patents



Source: Authors' own calculation using USPTO patent database and COMPUSTAT.

that became mega firms in year t but were not in year $t - 1$ (entrants); and S_t^{exit} refers to firms that were mega firms in year $t - 1$ but not in year t (exiters).

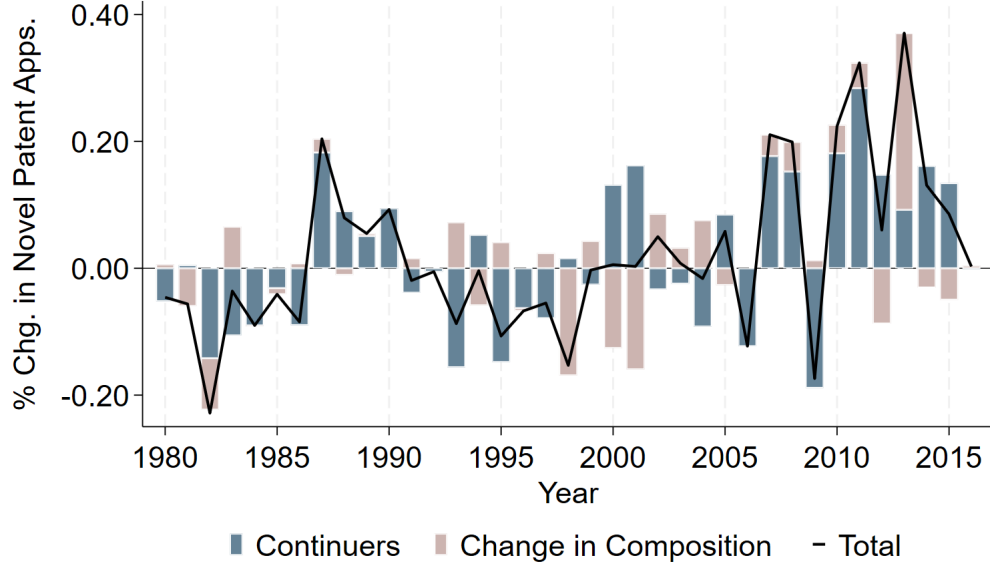
Dividing both sides by the average level of novel patenting, $\bar{N}_{m,t} = \frac{N_{m,t} + N_{m,t-1}}{2}$, and expressing each term as a weighted average, we obtain:

$$\frac{\Delta N_{m,t}}{\bar{N}_{m,t}} = \sum_{i \in S_t^{\text{cont}}} \frac{\Delta n_{i,t}}{\bar{n}_{i,t}} \cdot \frac{\bar{n}_{i,t}}{\bar{N}_{m,t}} + \sum_{i \in S_t^{\text{new}}} \frac{(n_{i,t} - 0)}{\bar{n}_{i,t}} \cdot \frac{\bar{n}_{i,t}}{\bar{N}_{m,t}} + \sum_{i \in S_t^{\text{exit}}} \frac{(0 - n_{i,t-1})}{\bar{n}_{i,t}} \cdot \frac{\bar{n}_{i,t}}{\bar{N}_{m,t}}.$$

Each term on the right-hand side represents a weighted average of firm-level growth rates in novel patent applications, where the weights are given by $\bar{n}_{i,t}/\bar{N}_{m,t}$. While the decomposition treats new mega firms as having zero patenting activity in year $t - 1$, this simply reflects their zero contribution to the *mega firm* group in that year. The same logic applies to exiters, whose contribution to the group becomes zero in year t .

Therefore, the growth rate of novel patent applications among mega firms can be ex-

Figure A8: Percent Change in Novel Patent Applications by Mega Firms:
Contribution by Continuers vs. Change in Composition



pressed as

$$g_t = g_t^{\text{cont}} + g_t^{\text{new}} + g_t^{\text{exit}} = g_t^{\text{cont}} + g_t^{\text{comp}},$$

where g_t^{cont} captures the contribution of continuing mega firms, and $g_t^{\text{comp}} (= g_t^{\text{new}} + g_t^{\text{exit}})$ captures the net effect of turnover. Note, by construction, $g_t^{\text{new}} \geq 0$ and $g_t^{\text{exit}} \leq 0$. Hence, for example, a positive g_t^{comp} indicates that novel patenting by entrants in year t exceeded that of exiters in year $t - 1$.

Figure A8 presents the decomposition graphically. The solid black line shows the overall growth rate g_t , while the blue bars represent the contribution from continuers (g_t^{cont}) and the rose bars show the contribution from turnover (g_t^{comp}). The results indicate that, for the vast majority of the sample period, year-to-year changes in novel patenting by mega firms were predominantly driven by continuers, rather than by changes in composition. A notable exception occurred around the year 2000, when new mega firms generated significantly fewer novel patents than the firms they replaced. Using the Shapley-Shorrocks variance decomposition, we find that g_t^{cont} accounts for 74% of the variation in g_t , while g_t^{comp} explains the remaining 26%.

A.9 Forward Citations and Self-citation Rates of Novel Patents

Tables A7 and A8 report results from the same regression specifications used in Tables 4 and 5, but with forward citations and self-forward-citation rates as alternative outcome variables. In these tables, “hits” are defined as novel patents that fall within the top one percentile of forward citations within their main CPC section over the first five years.

Table A7: Forward Citations on Novel Patents by Mega Firms

	(1)	(2)	(3)
	# Forward Citations	No Forward Citation	Hit
Mega firm	0.025 (0.020)	-0.008* (0.004)	0.001 (0.001)
Mega firm x Post 2001	0.225*** (0.074)	0.009 (0.007)	-0.001 (0.002)
Section x Year FE	Yes	Yes	Yes
Obs.	147129	147129	147129
Pseudo R-sq	0.09		
R-sq		0.08	0.00

Notes: Column (1) shows the result from a Poisson pseudo-maximum likelihood regression (PPML) with multi-way fixed effects, where the outcome variable is the number of forward citations received within five years. Columns (2) and (3) are results from linear probability OLS regressions. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Column (1) shows that, consistent with Table 4, novel patents filed by mega firms receive, on average, more forward citations after—and only after—2001. In contrast to Table 4, however, there is little difference between mega firms and non-mega firms in the extreme outcomes shown in Columns (2) and (3).

Table A8: Self-forward-citation Rates of Novel Patents by Mega Firms

	(1)	(2)	(3)	(4)
	Self-fwd.-cit. Rate	Self-fwd.-cit. Rate	Self-fwd.-cit. Rate	Self-fwd.-cit. Rate
Mega firm	0.065*** (0.004)	0.060*** (0.004)	0.134*** (0.028)	0.119*** (0.030)
Mega firm x Post 2001	0.004 (0.007)	0.009 (0.007)	0.009 (0.047)	-0.045 (0.049)
Post 2001	0.114*** (0.002)		0.117*** (0.015)	
Section x Year FE	No	Yes	No	Yes
Condition on Hits	No	No	Yes	Yes
Obs.	100894	100894	1700	1659
R-sq	0.03	0.07	0.06	0.30

Notes: All columns show the result from linear probability OLS regressions, where the outcome variable is the share of self-forward-citation received. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Column (1) indicates that self-forward-citation rates increased overall after 2001, consistent with the findings of Akcigit and Ates (2023). However, in line with Table 5, there is no evidence for a faster increase for novel patents filed by mega firms. As shown in the first rows, mega firms had higher self-forward-citation rates prior to 2001, suggesting slower knowledge diffusion from their novel innovations during this period.